Introduction to Analysis I

Lecture notes, Chapters 1–5 and Appendix

Contents

Chapter 1. Real Numbers and Limits of Sequences 3

1.1 Field and order axioms 3

1.2 The completeness axiom 4

1.3 Limits of sequences 6

1.4 Monotone sequences, limsup and liminf 7

1.5 Countable and uncountable sets 8

1.6 Selected exercises 9

Chapter 2. Topology of Euclidean Space 11

2.1 Euclidean space 11

2.2 Open and closed sets 11

2.3 The Bolzano–Weierstrass theorem and Cauchy sequences 12

2.4 Series 14

2.5 Compact sets 15

2.6 Connected sets 16

2.7 Selected exercises 17

Chapter 3. Continuous Functions 19

3.1 Limits and continuity 19

3.2 Continuity, compactness and connectedness 20

3.3 Uniform continuity 22

3.4 Monotone functions 23

3.5 Selected exercises 23

Chapter 4. Differentiable Functions 26

4.1 Differentiability 26

4.2 Mean value theorems **27**

4.3 Taylor's theorem **28**

4.4 Selected exercises **29**

Chapter 5. The Riemann–Stieltjes Integral **32**

5.1 The Riemann integral **32**

5.2 Integrable functions **33**

5.3 The fundamental theorem of calculus **34**

5.4 Functions of bounded variation **35**

5.5 The Riemann–Stieltjes integral **36**

Appendix A. The Number Systems **39**

A.1 Integers and rationals **39**

A.2 Real numbers as classes of Cauchy sequences **39**

A.3 Selected exercises **41**

Chapter 1. Real Numbers and Limits of Sequences

We take $\mathbb{R}$ as given by the field, order and completeness axioms; Appendix A constructs such a field. Book numbers are in brackets.

1.1 Field and order axioms

Motivation. Every inequality manipulation in the course reduces to the field and order rules, which $\mathbb{R}$ shares with $\mathbb{Q}$.

Definition (Ordered field)

A **field** is a set F with operations $+$ and $\cdot$, both associative and commutative, with $\cdot$ distributive over $+$, identities $0 \neq 1$, additive inverses $-a$, and multiplicative inverses a^{-1} for $a \neq 0$ (axioms (F1)–(F9)). A field is **ordered** if there is a subset P (the positive elements) with (O1) $a, b \in P \Rightarrow a+b \in P$; (O2) $a, b \in P \Rightarrow ab \in P$; (O3) for each a exactly one of $a \in P$, $a = 0$, $-a \in P$ holds. We write $a < b$ (or $b > a$) if $b - a \in P$, and $a \leq b$ if $a < b$ or $a = b$.

Proposition 1.1 (1.1.1)

In a field:
(a) $-(-a) = a$, and $(a^{-1})^{-1} = a$ for $a \neq 0$;
(b) $a + b = a + c \Rightarrow b = c$, and $ab = ac$, $a \neq 0 \Rightarrow b = c$;
(c) $ab = 0 \iff a = 0$ or $b = 0$;
(d) $a \cdot 0 = 0$, $(-a)b = -(ab)$, $(-a)(-b) = ab$;
(e) the identities and inverses are unique.

Proof. (e) If $0, 0'$ are additive identities, $0 = 0 + 0' = 0'$; if b, b' are inverses of a, $b = b + (a + b') = (b + a) + b' = b'$; likewise for $\cdot$.

(b) Add $-a$ to both sides (multiply by a^{-1}).

(a) a and $-(-a)$ are both additive inverses of $-a$; use (e).

(d) $a \cdot 0 + a \cdot 0 = a(0 + 0) = a \cdot 0 = a \cdot 0 + 0$, so $a \cdot 0 = 0$ by (b); $ab + (-a)b = (a - a)b = 0$ gives $(-a)b = -(ab)$, and then $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.

(c) If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ∎

Proposition 1.2 (1.1.2)

In an ordered field:
(a) $a \leq b$ and $b \leq a$ imply $a = b$;
(b) $a < b$, $b < c$ imply $a < c$;
(c) $a + b < a + c \iff b < c$;
(d) if $a > 0$ then $b < c \iff ab < ac$, and if $a < 0$ then $b < c \iff ab > ac$;
(e) $a^2 > 0$ for $a \neq 0$, in particular $1 > 0$;
(f) $a > 0 \Rightarrow a^{-1} > 0$, and $0 < a < b \Rightarrow 0 < b^{-1} < a^{-1}$;
(g) $a < b$, $c < d \Rightarrow a + c < b + d$;
(h) for $a, b \geq 0$: $a < b \iff a^2 < b^2$, and $a^2 = b^2 \Rightarrow a = b$.

Proof. (a) If $a \neq b$, both $b - a$ and $a - b = -(b - a)$ would be in P, against (O3).

(b) $c - a = (c - b) + (b - a) \in P$.

(c) $(a+c)-(a+b)=c-b$.

(d) $ac-ab=a(c-b)$; for $a>0$ use (O2) and trichotomy, for $a<0$ note $-a>0$.

(e) If $a>0$, $a^2\in P$; if $a<0$, $a^2=(-a)^2\in P$; and $1=1^2$.

(f) If $a^{-1}\le 0$ then $1=aa^{-1}\le 0$; multiply $a<b$ by $a^{-1}b^{-1}>0$.

(g) $(b+d)-(a+c)=(b-a)+(d-c)$.

(h) $b^2-a^2=(b-a)(b+a)$ and $b+a>0$ when $a,b\ge 0$ are not both 0. ■

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. $\mathbb{C}$ is not: in an ordered field $-1<0<1=1^2$, but $i^2=-1$ would have to be positive. No finite field is ordered, since $1, 1+1, 1+1+1,\dots$ are strictly increasing.

The **absolute value** is $|a|=a$ for $a\ge 0$ and $|a|=-a$ for $a<0$.

Proposition 1.3 (1.1.3)

(a) $|a|\ge 0$, and $|a|=0 \iff a=0$;
(b) $|ab|=|a||b|$;
(c) for $b\ge 0$, $|a|\le b \iff -b\le a\le b$;
(d) $|a+b|\le |a|+|b|$ (triangle inequality);
(e) $||a|-|b||\le |a-b|$.

Proof. (a), (b) are case checks.

(c) If $a\ge 0$, $|a|\le b \iff a\le b$, and $-b\le 0\le a$ automatically; similarly for $a<0$.

(d) From $-|a|\le a\le |a|$ and the same for b, $-(|a|+|b|)\le a+b\le |a|+|b|$; apply (c).

(e) $|a|=|(a-b)+b|\le |a-b|+|b|$, and symmetrically. ■

1.2 The completeness axiom

Motivation. $\{x\in\mathbb{Q}: x^2<2\}$ has no supremum in $\mathbb{Q}$; completeness removes such holes and drives every existence theorem in analysis.

$S\subset\mathbb{R}$ is **bounded above** if it has an **upper bound** u ($s\le u$ for all $s\in S$). A **least upper bound** (**supremum**) $\sup S$ is an upper bound smaller than every other upper bound; it is unique if it exists. Lower bounds and $\inf S$ are defined dually.

Definition (Completeness axiom)

(C) Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Proposition 1.4 (1.2.1)

In an ordered field the following are equivalent:
(a) every nonempty set bounded above has a supremum;
(b) every nonempty set bounded below has an infimum.

Proof. (a)⇒(b): let $S\ne\emptyset$ be bounded below and L the set of lower bounds of S; $L\ne\emptyset$, and every $s\in S$ is an upper bound of L. Let $a=\sup L$. Each $s\in S$ is an upper bound of L, so $a\le s$: $a\in L$. If $b\in L$ then $b\le a$. So $a=\inf S$.

(b)⇒(a) is symmetric, or apply (b) to $-S=\{-s\}$. ■

A useful reformulation: $u = \sup S$ iff u is an upper bound and for every $\epsilon > 0$ there is $s \in S$ with $s > u - \epsilon$.

Proposition 1.5 (1.2.2)

$\mathbb{N}$ is not bounded above.

Proof. Otherwise $a = \sup \mathbb{N}$ exists; $a - 1$ is not an upper bound, so $a - 1 < n$ for some $n \in \mathbb{N}$, and then $n + 1 > a$ with $n + 1 \in \mathbb{N}$, a contradiction. ■

Corollary 1.6 (1.2.3)

$\inf\{1/n : n \in \mathbb{N}\} = 0$; that is, for every $\epsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \epsilon$.

Proof. 0 is a lower bound. If $\epsilon > 0$ were a lower bound, then $n \leq 1/\epsilon$ for all n, contradicting Proposition 1.5. ■

Corollary 1.7 (1.2.4, Archimedean property)

For $a > 0$ and $b \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $na > b$.

Proof. By Proposition 1.5 choose $n > b/a$. ■

Theorem 1.8 (1.2.5, density of the rationals)

If $a < b$ there is $r \in \mathbb{Q}$ with $a < r < b$.

Proof. First let $0 \leq a < b$. Choose n with $1/n < b - a$ (Corollary 1.6). The set $\{m \in \mathbb{N} : m/n > a\}$ is nonempty (Corollary 1.7); let m be its least element. Then $(m-1)/n \leq a$ (if $m = 1$ this holds as $a \geq 0$), so $a < m/n \leq a + 1/n < b$. If $a < 0$, choose $k \in \mathbb{N}$ with $k > -a$, find r between $a + k \geq 0$ and $b + k$, and take $r - k$. ■

The irrationals are also dense: apply the theorem to $a/\sqrt{2} < b/\sqrt{2}$ (using Proposition 1.9).

Proposition 1.9 (1.2.6)

For each $a > 0$ there is a unique $b > 0$ with $b^2 = a$.

Proof. Uniqueness: Proposition 1.2(h).

Existence: $S = \{x > 0 : x^2 < a\} \ni t = a/(1 + a)$ (as $t < 1$, $t < a$ give $t^2 < t < a$), and S is bounded by $1+a$ ($x > 1+a \Rightarrow x^2 > x > a$). Let $b = \sup S > 0$. If $b^2 < a$: for $0 < h < \min\{1, (a-b^2)/(2b+1)\}$, $(b+h)^2 \leq b^2 + h(2b+1) < a$, so $b+h \in S$, a contradiction. If $b^2 > a$: for $0 < h < \min\{b, (b^2-a)/2b\}$, $(b-h)^2 > b^2 - 2bh > a$, so $b - h$ is an upper bound ($x \geq b - h \Rightarrow x^2 > a$), a contradiction. So $b^2 = a$. ■

In particular $\mathbb{Q}$ does not satisfy (C): $\{x \in \mathbb{Q} : x > 0, x^2 < 2\}$ has no rational supremum, as $\sqrt{2}$ is irrational.

Theorem 1.10 (1.2.7, uniqueness of the real field)

Let F and G be ordered fields satisfying the completeness axiom. There is a bijection $f : F \to G$ with $f(a+b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$, and $a < b \iff f(a) < f(b)$.

Proof. $n \mapsto 1 + \cdots + 1$, extended to $\mathbb{Z}$ and quotients, embeds $\mathbb{Q}$ in each ordered field preserving operations and order (Proposition 1.2); let $\phi : \mathbb{Q}_F \to \mathbb{Q}_G$ be the induced isomorphism. F, G are Archimedean (Corollary 1.7), so rationals are dense in each (Theorem 1.8). Define

$$f(x) = \sup\{\phi(r) : r \in \mathbb{Q}_F,\ r < x\}.$$

The set is nonempty and bounded above by $\phi(s)$ for rational $s > x$, so $f(x)$ exists.

Order: $x < r < s < y$ with r, s rational gives $f(x) \le \phi(r) < \phi(s) \le f(y)$; and $f(r) = \phi(r)$ for rational r (density). So f is strictly increasing, hence injective.

Onto: for $z \in G$ put $x = \sup\{r \in \mathbb{Q}_F : \phi(r) < z\}$; density in G and monotonicity give $f(x) = z$.

Additivity: rationals $r < x$, $s < y$ give $r + s < x + y$, so $\phi(r) + \phi(s) \le f(x + y)$ and $f(x) + f(y) \le f(x + y)$; applying this to $-x, -y$ (with $f(-x) = -f(x)$, by definition and density) gives equality.

Multiplicativity, $x, y > 0$: products rs of positive rationals $r < x$, $s < y$ are cofinal among rationals below xy, so $f(xy) = f(x)f(y)$; other signs via $f(-x) = -f(x)$. ■

So the axioms determine $\mathbb{R}$ up to isomorphism.

Proposition 1.11 (1.2.8)

For nonempty bounded $S, T \subset \mathbb{R}$, $\sup(S+T) = \sup S + \sup T$ and $\inf(S+T) = \inf S + \inf T$, where $S + T = \{s + t\}$. If S, T consist of positive numbers, $\sup(ST) = \sup S \cdot \sup T$ and $\inf(ST) = \inf S \cdot \inf T$.

Proof. $\sup S + \sup T$ is an upper bound of $S+T$. Given $\epsilon > 0$ choose $s > \sup S - \epsilon/2$, $t > \sup T - \epsilon/2$; then $s + t > \sup S + \sup T - \epsilon$. For products, $\sup S \sup T$ is an upper bound of ST; with $a = \sup S$, $b = \sup T$ and $0 < \delta < \min(a, b)$, choose $s > a - \delta$, $t > b - \delta$, so $st > (a-\delta)(b-\delta) = ab - \delta(a+b-\delta)$, which is arbitrarily close to ab. For infima, $\inf S \inf T$ is a lower bound of ST, and with $s < \inf S + \delta$, $t < \inf T + \delta$ we get $st < \inf S \inf T + \delta(\inf S + \inf T + \delta)$. ■

1.3 Limits of sequences

Motivation. The ε–N definition makes "arbitrarily close" provable; the limit laws then spare us ε each time.

(x_n) **converges** to x, $\lim x_n = x$, if for every $\epsilon > 0$ there is N with $|x_n - x| < \epsilon$ for all $n \ge N$.

Proposition 1.12 (1.3.1)

Limits are unique.

Proof. If $x_n \to x$, $x_n \to y$ and $x \ne y$, take $\epsilon = |x - y|/2$ and $n \ge \max(N_1, N_2)$: $|x - y| \le |x - x_n| + |x_n - y| < 2\epsilon = |x - y|$. ■

Proposition 1.13 (1.3.2)

Every convergent sequence is bounded.

Proof. If $x_n \to x$, choose N with $|x_n - x| < 1$ for $n \ge N$; then $|x_n| \le \max\{|x_1|, \ldots, |x_{N-1}|, |x| + 1\}$. ■

Theorem 1.14 (1.3.3, limit laws)

If $x_n \to x$ and $y_n \to y$, then (a) $x_n + y_n \to x + y$;
(b) $ax_n \to ax$;
(c) $x_n y_n \to xy$;
(d) if $y \neq 0$, then $y_n \neq 0$ for large n and $x_n/y_n \to x/y$;
(e) if $x_n \leq y_n$ for all large n, then $x \leq y$;
(f) (squeeze) if $x_n \leq z_n \leq y_n$ and $x = y$, then $z_n \to x$.

Proof. (a) $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y|$; use $\epsilon/2$ for each.

(b) is a special case of (c).

(c) With M bounding $|x_n|$ (Proposition 1.13), $|x_n y_n - xy| \leq |x_n||y_n - y| + |y||x_n - x| \leq M|y_n - y| + |y||x_n - x|$.

(d) For large n, $|y_n| > |y|/2$, and $|1/y_n - 1/y| = |y - y_n|/(|y_n||y|) \leq 2|y_n - y|/|y|^2$; combine with (c).

(e) If $x > y$, take $\epsilon = (x - y)/2$; for large n, $x_n > x - \epsilon = y + \epsilon > y_n$.

(f) For large n, $x - \epsilon < x_n \leq z_n \leq y_n < x + \epsilon$. ■

Example 1.1

(a) $1/n \to 0$ by Corollary 1.6.
(b) For $|r| < 1$, $r^n \to 0$: write $1/|r| = 1 + h$, $h > 0$; then $(1 + h)^n \geq 1 + nh$ (Bernoulli), so $|r|^n \leq 1/(1 + nh)$.
(c) $n^{1/n} \to 1$: with $n^{1/n} = 1 + h_n$, $n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)}$.

1.4 Monotone sequences, limsup and liminf

Motivation. Intrinsic convergence criteria: bounded monotone sequences converge, and lim sup, lim inf always exist.

Theorem 1.15 (1.4.1, monotone convergence)

A bounded monotone sequence of real numbers converges.

Proof. Let (x_n) be increasing and bounded, and $x = \sup\{x_n\}$ (axiom (C)). Given ϵ, some $x_N > x - \epsilon$, and then $x - \epsilon < x_N \leq x_n \leq x$ for $n \geq N$. For decreasing sequences use inf. ■

Corollary 1.16 (1.4.2)

A series of nonnegative terms whose partial sums are bounded converges.

Proof. The partial sums form a bounded increasing sequence. ■

Example 1.2

$(1 + 1/n)^n$ is increasing and bounded by 3 (binomial expansion), so it converges to a number e. $\sum 1/n^p$ converges for $p > 1$: grouping terms in blocks $2^k \leq n < 2^{k+1}$, the partial sums are at most $\sum_k 2^{k(1-p)} < \infty$. $\sum 1/n$ diverges: the block sums are at least $\frac{1}{2}$.

For a bounded sequence (x_n), the sequences $\sup_{k\ge n} x_k$ and $\inf_{k\ge n} x_k$ are decreasing and increasing, so they converge (Theorem 1.15); define

$$\limsup_n x_n = \lim_n \sup_{k\ge n} x_k = \inf_n \sup_{k\ge n} x_k, \qquad \liminf_n x_n = \lim_n \inf_{k\ge n} x_k = \sup_n \inf_{k\ge n} x_k.$$

Clearly $\liminf x_n \le \limsup x_n$, and $\liminf x_n = -\limsup(-x_n)$.

Proposition 1.17 (1.4.3)

For bounded (x_n), $a = \limsup x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n < a + \epsilon$ for all $n \ge N$, and (b) for every $\epsilon > 0$, $x_n > a - \epsilon$ for infinitely many n.

Proof. Let $s_n = \sup_{k\ge n} x_k \downarrow a$.

(a) Choose N with $s_N < a + \epsilon$; then $x_n \le s_N$ for $n \ge N$.

(b) For each N, $s_N \ge a > a - \epsilon$, so some $k \ge N$ has $x_k > a - \epsilon$. Conversely, if a satisfies (a) and (b), then by (a) $s_n \le a + \epsilon$ for large n, so $\limsup \le a + \epsilon$; by (b) $s_n > a - \epsilon$ for all n, so $\limsup \ge a - \epsilon$. As ϵ is arbitrary, $\limsup x_n = a$. ∎

Proposition 1.18 (1.4.4)

For bounded (x_n), $a = \liminf x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n > a - \epsilon$ for all $n \ge N$, and (b) for every $\epsilon > 0$, $x_n < a + \epsilon$ for infinitely many n.

Proof. Apply Proposition 1.17 to $(-x_n)$. ∎

Theorem 1.19 (1.4.5)

If $x_n \to a$, then $\limsup x_n = \liminf x_n = a$. Conversely, if (x_n) is bounded and $\limsup x_n = \liminf x_n = a$, then $x_n \to a$.

Proof. If $x_n \to a$, then a satisfies conditions (a) and (b) of both propositions. Conversely, part (a) of Propositions 1.17 and 1.18 give $a - \epsilon < x_n < a + \epsilon$ for all large n. ∎

Theorem 1.20 (1.4.6, nested interval theorem)

If $I_n = [a_n, b_n]$ are bounded closed intervals with $I_{n+1} \subset I_n$ for all n, then $\bigcap_n I_n \ne \emptyset$. If moreover $b_n - a_n \to 0$, the intersection is a single point.

Proof. (a_n) is increasing and bounded by b_1, so $a_n \to a = \sup a_n$; for all m, n, $a_n \le a_{\max(m,n)} \le b_{\max(m,n)} \le b_m$, so $a \le b_m$. Thus $a_m \le a \le b_m$ for all m. If x, y both lie in every I_n, $|x - y| \le b_n - a_n \to 0$. ∎

Conversely, nested intervals plus the Archimedean property give (C): bisect $[s_0, u_0]$ ($s_0 \in S$, u_0 an upper bound), keeping the half whose right end is an upper bound and which meets S; the common point is $\sup S$. Theorem 1.15 is likewise equivalent to (C).

1.5 Countable and uncountable sets

Motivation. Countability makes precise that there are "more" reals than rationals.

A set is **countable** if it is finite or in bijection with $\mathbb{N}$.

Proposition 1.21 (countable unions)

Subsets of countable sets are countable; $\mathbb{N} \times \mathbb{N}$ is countable; a countable union of countable sets is countable. Hence $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof. Infinite $A \subset \mathbb{N}$: $a_1 = \min A$, $a_{k+1} = \min(A - \{a_1, \dots, a_k\})$ enumerates it. $(m, n) \mapsto 2^m 3^n$ injects $\mathbb{N} \times \mathbb{N}$ into $\mathbb{N}$. With surjections $f_k : \mathbb{N} \to A_k$ $(A_k \neq \emptyset)$, $(k, n) \mapsto f_k(n)$ maps $\mathbb{N} \times \mathbb{N}$ onto $\bigcup A_k$, and an image of a countable set is countable (least preimages). $\mathbb{Q}$ is the image of $\mathbb{Z} \times \mathbb{N}$ under $(p, q) \mapsto p/q$. ■

Theorem 1.22 (Cantor)

$\mathbb{R}$ is uncountable.

Proof. Let $x_1, x_2, \dots$ be any sequence of reals. Choose a closed interval I_1 of positive length not containing x_1, then $I_2 \subset I_1$ not containing x_2, and so on. By Theorem 1.20 some $x \in \bigcap I_n$, and $x \neq x_n$ for every n. ■

Consequently the irrational numbers are uncountable, and every interval of positive length is uncountable.

1.6 Selected exercises

Motivation. Three book exercises: a series tail estimate, coupled monotone sequences, and countability.

Exercise 1.6.10 (e is irrational)

Let $s_n = \sum_{k=0}^{n} \frac{1}{k!}$.
(a) Show that (s_n) converges; call its limit e. (It agrees with $\lim(1 + 1/n)^n$ of Example 1.2, which is not needed here.)
(b) Show that $0 < e - s_n < \dfrac{1}{n!\, n}$ for every $n \geq 1$.
(c) Show that e is irrational.

Solution.

(a) and (b).

Step 1. For $k \geq n+1$ we have $k! = (n+1)!\,(n+2)(n+3)\cdots k \geq (n+1)!\,(n+2)^{k-n-1}$. Hence for $m > n$,

$$0 < s_m - s_n = \sum_{k=n+1}^{m} \frac{1}{k!} \leq \frac{1}{(n+1)!} \sum_{j=0}^{\infty} \frac{1}{(n+2)^j} = \frac{1}{(n+1)!} \cdot \frac{n+2}{n+1}.$$

Step 2. $n = 1$: $s_m \leq s_1 + \frac{1}{2} \cdot \frac{3}{2} = \frac{11}{4}$. So (s_m) is increasing and bounded, hence convergent (Theorem 1.15).

Step 3. Let $m \to \infty$ in Step 1, using $s_m - s_n \geq s_{n+1} - s_n$:

$$0 < \frac{1}{(n+1)!} \leq e - s_n \leq \frac{n+2}{(n+1)!\,(n+1)} = \frac{1}{n!\, n} \cdot \frac{n(n+2)}{(n+1)^2} < \frac{1}{n!\, n},$$

since $n(n+2) = (n+1)^2 - 1$.

(c)

Step 1. Suppose $e = p/q$, $p, q \in \mathbb{N}$ ($e > s_1 = 2 > 0$). Then $q!\,e = p\,(q-1)!$ and $q!\,s_q = \sum_{k=0}^{q} q!/k!$ are integers ($k! \mid q!$ for $k \le q$).

Step 2. So $q!(e - s_q) \in \mathbb{Z}$, while $0 < q!(e - s_q) < 1/q \le 1$ by (b), a contradiction. Hence $e \notin \mathbb{Q}$. ■

Exercise 1.6.15 (Arithmetic–geometric mean)

Let $a_1 \ge b_1 > 0$ and define recursively

$$a_{n+1} = \frac{a_n + b_n}{2}, \qquad b_{n+1} = \sqrt{a_n b_n}.$$

Show that (a_n) and (b_n) converge to the same limit.

Solution. If $a_1 = b_1$ both sequences are constant; let $a_1 > b_1$ (square roots: Proposition 1.9).

Step 1. For $x > y > 0$, $\frac{x+y}{2} - \sqrt{xy} = \frac{1}{2}(\sqrt{x} - \sqrt{y})^2 > 0$; by induction $a_n > b_n > 0$.

Step 2. Then $a_{n+1} = \frac{a_n + b_n}{2} < a_n$ and $b_{n+1} = \sqrt{a_n b_n} > b_n$, so

$$b_1 < b_2 < \cdots < b_n < a_n < \cdots < a_2 < a_1.$$

Both are monotone and bounded, so $a_n \to L$, $b_n \to M$ (Theorem 1.15).

Step 3. Since $b_{n+1} > b_n$,

$$0 < a_{n+1} - b_{n+1} < a_{n+1} - b_n = \frac{a_n - b_n}{2},$$

so $0 < a_n - b_n < (a_1 - b_1)/2^{n-1} \to 0$. By the limit laws (Theorem 1.14), $L - M = \lim(a_n - b_n) = 0$. ■

Exercise 1.6.17 (Algebraic numbers)

A real number is **algebraic** if it is a root of a nonzero polynomial with integer coefficients, and **transcendental** otherwise.
(a) Show that the set of algebraic numbers is countable.
(b) Deduce that there are uncountably many transcendental numbers.

Solution.

(a)

Step 1. $A \times B = \bigcup_{a \in A} \{a\} \times B$ is countable for countable A, B (Proposition 1.21); by induction $\mathbb{Z}^{d+1}$ is countable.

Step 2. Nonzero integer polynomials of degree $\le d$ form a set P_d injecting into $\mathbb{Z}^{d+1}$ via $\sum c_k x^k \mapsto (c_0, \ldots, c_d)$; so P_d and $P = \bigcup_d P_d$ are countable.

Step 3. A nonzero p of degree d has at most d real roots: none if $d = 0$; if $d \ge 1$ and $p(r) = 0$, then $p(x) = (x - r)q(x)$ with $\deg q = d - 1$, and any root $s \ne r$ is a root of q (Proposition 1.1(c)); induct.

Step 4. The algebraic numbers $\bigcup_{p \in P} \{x : p(x) = 0\}$ form a countable union of finite sets, hence are countable (Proposition 1.21).

(b) If the transcendentals T were countable, so would be $\mathbb{R} = T \cup (\mathbb{R} - T)$, contradicting Theorem 1.22. So T is uncountable. ■

Chapter 2. Topology of Euclidean Space

2.1 Euclidean space

Motivation. Cauchy–Schwarz makes the Euclidean distance on $\mathbb{R}^n$ satisfy the triangle inequality.

$\mathbb{R}^n$ is the set of n-tuples $x = (x_1, \dots, x_n)$ with coordinatewise addition and scalar multiplication, the **inner product** $\langle x, y\rangle = \sum x_i y_i$, the **norm** $\|x\| = \langle x, x\rangle^{1/2}$ and the **distance** $\|x - y\|$.

Theorem 2.1 (2.1.1, Cauchy–Schwarz inequality)

$|\langle x, y\rangle| \le \|x\|\|y\|$ for all $x, y \in \mathbb{R}^n$, with equality iff x, y are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $\|x - z\| \le \|x - y\| + \|y - z\|$.

Proof. If $y = 0$ both sides vanish. Otherwise, for real t, $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$; take $t = \langle x, y\rangle / \|y\|^2$ to get $0 \le \|x\|^2 - \langle x, y\rangle^2/\|y\|^2$. Equality forces $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ■

Also $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, so a sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges.

2.2 Open and closed sets

Motivation. Open sets suit differentiation, closed sets (containing their limits) guarantee maxima.

The **ϵ-neighbourhood** of x is $N(x, \epsilon) = \{y : \|y - x\| < \epsilon\}$. A point x is an **interior point** of A if some $N(x, \epsilon) \subset A$; A is **open** if all its points are interior. x is a **limit point** of A if every $N(x, \epsilon)$ contains a point of A other than x; A' denotes the set of limit points, and $\overline{A} = A \cup A'$ is the **closure**. A is **closed** if $\mathbb{R}^n - A$ is open. Points of $A - A'$ are **isolated points**. Every $N(x, \epsilon)$ is open, by the triangle inequality.

Proposition 2.2 (2.2.1)

(a) $\emptyset$ and $\mathbb{R}^n$ are open;
(b) unions of open sets are open;
(c) finite intersections of open sets are open. Dually, $\emptyset, \mathbb{R}^n$ are closed, intersections of closed sets are closed, and finite unions of closed sets are closed.

Proof. (a), (b) are immediate.

(c) If $x \in U_1 \cap \cdots \cap U_k$ with $N(x, \epsilon_i) \subset U_i$, then $N(x, \min \epsilon_i)$ lies in the intersection. The dual statements follow by De Morgan's laws. ■

An infinite intersection of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.

Proposition 2.3 (2.2.2)

For $A \subset \mathbb{R}^n$ and $x \in \mathbb{R}^n$:
(a) $x \in A'$ iff some sequence in $A - \{x\}$ converges to x;
(b) $x \in \overline{A}$ iff some sequence in A converges to x.

Proof. (a) If $x \in A'$, pick $x_k \in A \cap N(x, 1/k)$, $x_k \ne x$. Conversely, if $x_k \to x$ with $x_k \in A - \{x\}$, every $N(x, \epsilon)$ contains some x_k.

(b) If $x \in A$ use the constant sequence; otherwise $x \in A'$ and use (a). Conversely, if $x_k \to x$, $x_k \in A$, either $x = x_k$ for some k or the x_k lie in $A - \{x\}$. ■

Theorem 2.4 (2.2.3)

For $F \subset \mathbb{R}^n$ the following are equivalent:
(a) F is closed;
(b) $F' \subset F$;
(c) $F = \overline{F}$.

Proof. (a)⇒(b): if $x \notin F$, some $N(x, \epsilon)$ misses F, so $x \notin F'$. (b)⇒(c): $\overline{F} = F \cup F' = F$. (c)⇒(a): if $x \notin F$, then $x \notin F'$, so some $N(x, \epsilon)$ contains no point of F (other than x, which is not in F); thus $\mathbb{R}^n - F$ is open. ■

Corollary 2.5 (2.2.4)

A' is closed for every $A \subset \mathbb{R}^n$; hence $\overline{A}$ is closed, and it is the smallest closed set containing A.

Proof. Let $x \in (A')'$ and $\epsilon > 0$. Choose $y \in A'$ with $0 < \|y - x\| < \epsilon$, and $\delta = \min(\|y - x\|, \epsilon - \|y - x\|) > 0$. Then $N(y, \delta)$ contains a point $z \in A$ with $z \neq y$; $z \in N(x, \epsilon)$, and $z \neq x$ because $\|z - y\| < \|x - y\|$. So $x \in A'$. The same argument shows $(\overline{A})' \subset A'$: if $N(x, \epsilon)$ contains $y \in A \cup A'$ with $y \neq x$, either $y \in A$, or $y \in A'$ and a small neighbourhood of y inside $N(x, \epsilon)$ contains a point of A other than x. So $\overline{A}$ is closed by Theorem 2.4. If $F \supset A$ is closed, $F \supset F' \supset A'$, so $F \supset \overline{A}$. ■

For $X \subset \mathbb{R}^n$, a set $U \subset X$ is **open in** X (relatively open) if for each $x \in U$ there is $\epsilon > 0$ with $N(x, \epsilon) \cap X \subset U$.

Proposition 2.6 (2.2.5)

For $U \subset X \subset \mathbb{R}^n$ the following are equivalent:
(a) U is open in X;
(b) $U = V \cap X$ for some open $V \subset \mathbb{R}^n$.

Proof. (a)⇒(b): for $x \in U$ choose ϵ_x with $N(x, \epsilon_x) \cap X \subset U$, and let $V = \bigcup_x N(x, \epsilon_x)$. (b)⇒(a): if $x \in V \cap X$ and $N(x, \epsilon) \subset V$, then $N(x, \epsilon) \cap X \subset U$. ■

Similarly, $F \subset X$ is closed in X iff $F = E \cap X$ with E closed in $\mathbb{R}^n$.

2.3 The Bolzano–Weierstrass theorem and Cauchy sequences

Motivation. Bounded sequences have convergent subsequences, which yields the Cauchy criterion: convergence without knowing the limit.

A **box** is $B = I_1 \times \cdots \times I_n$ with bounded closed intervals I_k.

Lemma 2.7 (2.3.1, nested boxes)

If $B_1 \supset B_2 \supset \cdots$ are boxes, then $\bigcap_m B_m \neq \emptyset$.

Proof. Write $B_m = I_1^m \times \cdots \times I_n^m$. For each k, the intervals I_k^m are nested, so contain a common point x_k (Theorem 1.20); then $(x_1, \ldots, x_n) \in \bigcap B_m$. ■

Theorem 2.8 (2.3.2, Bolzano–Weierstrass)

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point.

Proof. A lies in a box B_1. Divide B_1 into 2^n boxes by bisecting each side; one of them, B_2, contains infinitely many points of A. Repeating, we obtain nested boxes B_m whose sides have length 2^{1-m} times those of B_1, each containing infinitely many points of A. Let $x \in \bigcap B_m$ (Lemma 2.7). Any $N(x,\epsilon)$ contains B_m for large m (its diameter tends to 0), hence infinitely many points of A; so $x \in A'$. ■

Proposition 2.9 (2.3.3)

If x is a limit point of A, then $A \cap N(x,\epsilon)$ is infinite for every $\epsilon > 0$.

Proof. If $A \cap N(x,\epsilon) - \{x\} = \{a_1, \ldots, a_k\}$, then $N(x,\delta)$ with $\delta = \min\|a_i - x\|$ contains no point of A other than x, a contradiction. ■

Theorem 2.10 (2.3.4)

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. If the range $\{x_k\}$ is finite, some value is taken infinitely often, giving a constant subsequence. Otherwise it is a bounded infinite set with a limit point x (Theorem 2.8). Each $N(x,1/j)$ contains infinitely many terms (Proposition 2.9), so we can choose $k_1 < k_2 < \cdots$ with $\|x_{k_j} - x\| < 1/j$. ■

For a bounded real sequence (a_n) let A be the set of limits of its convergent subsequences.

Theorem 2.11 (2.3.5)

$\limsup a_n = \max A$ and $\liminf a_n = \min A$.

Proof. Let $a = \limsup a_n$. By Proposition 1.17(b) we may choose $n_1 < n_2 < \cdots$ with $a_{n_k} > a - 1/k$, and by (a) of the same proposition $a_{n_k} < a + 1/k$ for large k; so $a_{n_k} \to a$ and $a \in A$. If $a_{n_k} \to b$, then $a_{n_k} < a + \epsilon$ for large k, so $b \le a$. The $\liminf$ case is symmetric. ■

A sequence is **Cauchy** if for every $\epsilon > 0$ there is N with $\|x_m - x_n\| < \epsilon$ for all $m, n \ge N$.

Proposition 2.12 (2.3.6)

Every convergent sequence is Cauchy, and every Cauchy sequence is bounded.

Proof. If $x_n \to x$, choose N with $\|x_n - x\| < \epsilon/2$ for $n \ge N$; then $\|x_m - x_n\| < \epsilon$. If (x_n) is Cauchy, choose N for $\epsilon = 1$; then $\|x_n\| \le \max\{\|x_1\|, \ldots, \|x_N\|\} + 1$. ■

Lemma 2.13 (2.3.7)

If a subsequence of a Cauchy sequence (x_n) converges to x, then (x_n) converges to x.

Proof. Given ϵ, choose N with $\|x_m - x_n\| < \epsilon/2$ for $m, n \ge N$, and k with $n_k \ge N$ and $\|x_{n_k} - x\| < \epsilon/2$. For $n \ge N$, $\|x_n - x\| \le \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \epsilon$. ■

Theorem 2.14 (2.3.8, Cauchy criterion)

Every Cauchy sequence in $\mathbb{R}^n$ converges.

Proof. It is bounded (Proposition 2.12), so has a convergent subsequence (Theorem 2.10); apply Lemma 2.13. ■

Corollary 2.15 (2.3.9, Cauchy criterion for series)

$\sum a_n$ converges iff for every $\epsilon > 0$ there is N with $|a_{m+1} + \cdots + a_n| < \epsilon$ for all $n > m \geq N$.

Proof. Apply Theorem 2.14 to the partial sums s_n, since $s_n - s_m = a_{m+1} + \cdots + a_n$. ■

In particular, if $\sum a_n$ converges then $a_n \to 0$, and if $\sum |a_n|$ converges (**absolute convergence**) then $\sum a_n$ converges, because $|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n|$.

2.4 Series

Motivation. Series are limits of partial sums; the tests below are practical forms of the Cauchy criterion.

Proposition 2.16 (2.4.1, rearrangement of nonnegative series)

Let $a_n \geq 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ a bijection. If $\sum a_n = s$, then $\sum a_{\tau(n)} = s$; if $\sum a_n$ diverges, so does $\sum a_{\tau(n)}$.

Proof. Each partial sum $\sum_{n \leq N} a_{\tau(n)}$ is at most $\sum_{n \leq M} a_n \leq s$ with $M = \max \tau(\{1, \ldots, N\})$, so $t = \sum a_{\tau(n)} \leq s$ by Corollary 1.16. Applying this to τ^{-1} gives $s \leq t$. The divergence statement follows by symmetry. ■

Absolutely convergent series may be rearranged freely (apply the proposition to $a_n^{\pm} = \max(\pm a_n, 0)$); conditionally convergent ones can be rearranged to any sum (Riemann).

Theorem 2.17 (2.4.2, alternating series test)

If $x_1 \geq x_2 \geq \cdots \geq 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and its sum s satisfies $|s - s_n| \leq x_{n+1}$.

Proof. $s_{2m} = (x_1 - x_2) + \cdots + (x_{2m-1} - x_{2m})$ increases, $s_{2m+1} = x_1 - (x_2 - x_3) - \ldots$ decreases, and $s_{2m} \leq s_{2m+1}$. Both converge (Theorem 1.15), and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$, so they have the same limit s, which lies between consecutive partial sums. ■

Theorem 2.18 (2.4.3, comparison test)

If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.

Proof. The partial sums of $\sum a_n$ are bounded by $\sum b_n$; use Corollary 1.16. ■

Proposition 2.19 (2.4.4, root test)

Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, $\sum a_n$ diverges.

Proof. If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large n (Proposition 1.17), and $\sum \beta^n < \infty$. If $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$. ■

Proposition 2.20 (2.4.5, ratio test)

Let $a_n \neq 0$. If $\limsup|a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely; if $|a_{n+1}/a_n| \geq 1$ for all large n, then $\sum a_n$ diverges.

Proof. If $|a_{n+1}/a_n| \leq \beta < 1$ for $n \geq N$, then $|a_n| \leq |a_N|\beta^{n-N}$. In the second case $|a_n|$ is eventually nondecreasing and positive, so $a_n \not\to 0$. ■

Proposition 2.21 (2.4.6)

If $c_n > 0$, then $\liminf \frac{c_{n+1}}{c_n} \leq \liminf c_n^{1/n} \leq \limsup c_n^{1/n} \leq \limsup \frac{c_{n+1}}{c_n}$.

Proof. Let $\beta = \limsup c_{n+1}/c_n$, assumed finite, and $\gamma > \beta$. For $n \geq N$, $c_{n+1} < \gamma c_n$, so $c_n < c_N\gamma^{-N}\gamma^n$ and $c_n^{1/n} < (c_N\gamma^{-N})^{1/n}\gamma \to \gamma$. Hence $\limsup c_n^{1/n} \leq \gamma$ for all $\gamma > \beta$. The left inequality is analogous. ■

So the root test is stronger than the ratio test: whenever the ratio test decides, so does the root test.

2.5 Compact sets

Motivation. Compactness turns local estimates into global ones; in $\mathbb{R}^n$ it means closed and bounded.

An **open cover** of K is a family of open sets whose union contains K; K is **compact** if every open cover has a finite subcover.

Proposition 2.22 (2.5.1)

Every compact set is bounded and closed.

Proof. The sets $N(0,k)$ cover K; finitely many do, so $K \subset N(0,m)$. If $x \notin K$, the open sets $U_k = \{y : \|y - x\| > 1/k\}$ cover K; a finite subcover gives $K \subset U_m$, so $N(x, 1/m)$ misses K. Hence $\mathbb{R}^n - K$ is open. ■

Theorem 2.23 (2.5.2)

Every box $B = I_1 \times \cdots \times I_n$ is compact.

Proof. Suppose an open cover $\{U_i\}$ of B has no finite subcover. Bisecting all sides, one of the 2^n sub-boxes, B_2, is not covered by finitely many U_i; continuing gives nested boxes B_m of diameter $\operatorname{diam} B_1/2^{m-1}$ none finitely covered. Let $x \in \bigcap B_m$ (Lemma 2.7) and $x \in U_i$, $N(x,\epsilon) \subset U_i$. For large m, $B_m \subset N(x,\epsilon) \subset U_i$, a contradiction. ■

Proposition 2.24 (2.5.3)

A closed subset F of a compact set K is compact.

Proof. If $\{U_i\}$ covers F, then $\{U_i\} \cup \{\mathbb{R}^n - F\}$ covers K; a finite subcover, with $\mathbb{R}^n - F$ removed, covers F. ■

Theorem 2.25 (2.5.4, Heine–Borel)

For $K \subset \mathbb{R}^n$ the following are equivalent:
(a) K is compact;
(b) K is bounded and closed;
(c) every infinite subset of K has a limit point in K.

Proof. (a)⇒(b): Proposition 2.22. (b)⇒(a): K lies in a box, which is compact (Theorem 2.23); apply Proposition 2.24. (b)⇒(c): Bolzano–Weierstrass gives a limit point, which is in K as K is closed. (c)⇒(b): if K were unbounded, choose $x_k \in K$ with $\|x_k\| > k$; the infinite set $\{x_k\}$ has no limit point at all. If $x \in K'$, Proposition 2.9 gives distinct $x_k \in K$ with $x_k \to x$; the set $\{x_k\}$ has x as its only limit point, so $x \in K$ by (c), and K is closed. ■

So K is compact iff every sequence in K has a subsequence converging in K.

Theorem 2.26 (2.5.5, finite intersection property)

If $\{K_i : i \in I\}$ are compact sets such that $\bigcap_{i\in J} K_i \neq \emptyset$ for every finite $J \subset I$, then $\bigcap_{i\in I} K_i \neq \emptyset$.

Proof. Fix i_0. If the intersection were empty, the open sets $\mathbb{R}^n - K_i$ would cover K_{i_0}, and finitely many $\mathbb{R}^n - K_{i_1}, \dots, \mathbb{R}^n - K_{i_m}$ would cover it; then $K_{i_0} \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

In particular a decreasing sequence of nonempty compact sets has nonempty intersection.

Corollary 2.27 (2.5.6)

A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point.

Proof. Let $S = \{s_1, s_2, \dots\}$ have no isolated point. Take $K_1 = \overline{N(y_1, r_1)}$, $y_1 \in S$. Inductively, $N(y_k, r_k) \cap S$ is infinite, so choose $y_{k+1} \in S \cap N(y_k, r_k)$, $y_{k+1} \neq s_k$, and a closed ball $K_{k+1} \subset N(y_k, r_k)$ about y_{k+1} with $s_k \notin K_{k+1}$. The $K_k \cap S$ are nonempty, compact, decreasing, so share a point (Theorem 2.26), which is in S yet differs from every s_k. ■

Thus a nonempty closed set without isolated points (a **perfect set**), such as the Cantor set, is uncountable.

2.6 Connected sets

Motivation. Intervals cannot be split into two separated pieces; this underlies the intermediate value theorem.

$S \subset \mathbb{R}^n$ is **disconnected** if there are open U, V with

$$U \cap S \neq \emptyset, \quad V \cap S \neq \emptyset, \quad U \cap V \cap S = \emptyset, \quad S \subset U \cup V,$$

and **connected** otherwise. A set $S \subset \mathbb{R}$ is an **interval** if $x, y \in S$ and $x < z < y$ imply $z \in S$.

Proposition 2.28 (2.6.1)

A subset S of $\mathbb{R}$ is connected iff it is an interval.

Proof. If $x < z < y$ with $x, y \in S$, $z \notin S$, then $(-\infty, z)$, (z, ∞) disconnect S. Conversely, let S be an interval disconnected by U, V, $x \in U \cap S$, $y \in V \cap S$, $x < y$; so $[x, y] \subset S$. Let $c = \sup([x, y] \cap U) \in S \subset U \cup V$. If $c \in U$: $c < y$ and $[c, c + \delta) \subset U$, contradicting the definition of c. If $c \in V$: $c > x$ and $(c - \delta, c] \subset V$ misses U, so c is not the supremum. Contradiction either way. ■

Proposition 2.29 (2.6.2)

If $\{C_i\}$ are connected and $\bigcap C_i \neq \emptyset$, then $C = \bigcup C_i$ is connected.

Proof. Suppose U, V disconnect C, and let $p \in \bigcap C_i$, say $p \in U$. Each C_i is connected and meets U (at p), so $C_i \cap V = \emptyset$ (otherwise U, V would disconnect C_i). Then $C \cap V = \emptyset$, a contradiction. ■

Convex sets, in particular balls and boxes, are connected: they are unions of segments through a common point, each segment being connected as a continuous image of an interval (Theorem 3.13).

2.7 Selected exercises

Motivation. Three book exercises: diameter and the Cauchy condition, and two facts on connectedness.

For a nonempty $X \subset \mathbb{R}^n$ let $\operatorname{diam} X = \sup\{\|x - y\| : x, y \in X\} \in [0, \infty]$.

Exercise 2.7.23 (Diameter)

(a) Show that $\operatorname{diam} \overline{X} = \operatorname{diam} X$ for every nonempty $X \subset \mathbb{R}^n$.
(b) Show that a sequence (x_k) in $\mathbb{R}^n$ is Cauchy iff $\lim_{N\to\infty} \operatorname{diam}\{x_k : k \geq N\} = 0$.
(c) Let $X_1 \supset X_2 \supset \cdots$ be nonempty closed bounded sets with $\operatorname{diam} X_k \to 0$. Show that $\bigcap_k X_k$ consists of exactly one point.

Solution.

(a)

Step 1. $X \subset \overline{X}$ gives $\operatorname{diam} X \leq \operatorname{diam} \overline{X}$; assume $\operatorname{diam} X < \infty$.

Step 2. For $x', y' \in \overline{X}$, $\epsilon > 0$, pick $x, y \in X$ within ϵ of x', y' (Proposition 2.3(b)):

$$\|x' - y'\| \leq \|x' - x\| + \|x - y\| + \|y - y'\| < \operatorname{diam} X + 2\epsilon.$$

so $\|x' - y'\| \leq \operatorname{diam} X$ and $\operatorname{diam} \overline{X} \leq \operatorname{diam} X$.

(b) $d_N = \operatorname{diam}\{x_k : k \geq N\}$ decreases in N.

Step 1. Cauchy: $\|x_m - x_k\| < \epsilon/2$ for $m, k \geq N$ gives $d_M \leq d_N \leq \epsilon/2$ for $M \geq N$, so $d_N \to 0$.

Step 2. Conversely, $d_N < \epsilon$ gives $\|x_m - x_k\| \leq d_N < \epsilon$ for $m, k \geq N$.

(c)

Step 1. The X_k are compact (Theorem 2.25), nonempty and decreasing, so $\bigcap_k X_k \neq \emptyset$ (Theorem 2.26).

Step 2. $x, y \in \bigcap_k X_k$ gives $\|x - y\| \leq \operatorname{diam} X_k \to 0$, so $x = y$.

Uniqueness needs $\operatorname{diam} X_k \to 0$ ($X_k = [0, 1]$); existence needs closedness ($X_k = (0, 1/k)$). ■

Exercise 2.7.25 (Sets between a connected set and its closure)

Let $A \subset \mathbb{R}^n$ be connected and $A \subset B \subset \overline{A}$. Show that B is connected. In particular $\overline{A}$ is connected.

Solution.

Step 1. Suppose open U, V disconnect B: $U \cap B \neq \emptyset$, $V \cap B \neq \emptyset$, $U \cap V \cap B = \emptyset$, $B \subset U \cup V$.

Step 2. As $A \subset B$, $U \cap V \cap A = \emptyset$ and $A \subset U \cup V$; A connected forces, say, $U \cap A = \emptyset$.

Step 3. Pick $x \in U \cap B$ and $N(x, \epsilon) \subset U$. As $x \in \overline{A}$, $N(x, \epsilon)$ contains a point of A (Proposition 2.3(b)), contradicting $U \cap A = \emptyset$. ■

Exercise 2.7.26 (Connected sets have no isolated points)

Let $A \subset \mathbb{R}^n$ be connected with at least two points. Show that every point of A is a limit point of A.

Solution.

Step 1. Suppose $x \in A$ is not a limit point of A: there is $\epsilon > 0$ with $N(x, 2\epsilon) \cap A = \{x\}$. Let

$$U = N(x, \epsilon), \qquad V = \{y \in \mathbb{R}^n : \|y - x\| > \epsilon\}.$$

Both are open ($N(y, \|y - x\| - \epsilon) \subset V$ for $y \in V$) and disjoint.

Step 2. $x \in U \cap A$; any $z \in A - \{x\}$ has $\|z - x\| \geq 2\epsilon$, so $z \in V \cap A$.

Step 3. Every $z \in A$ lies in $U \cup V$: either $z = x \in U$, or $\|z - x\| \geq 2\epsilon$ and $z \in V$. So U, V disconnect A, a contradiction. ■

Chapter 3. Continuous Functions

Throughout, $X \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^m$, and functions $f : X \to Y$.

3.1 Limits and continuity

Motivation. The ϵ–δ and sequential definitions of continuity agree, so Chapters 1–2 apply to functions.

Let x_0 be a limit point of X. We write $\lim_{x\to x_0} f(x) = y_0$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - y_0\| < \epsilon$ whenever $x \in X$ and $0 < \|x - x_0\| < \delta$. f is **continuous at** $x_0 \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(x_0)\| < \epsilon$ whenever $x \in X$, $\|x - x_0\| < \delta$; equivalently, x_0 is isolated in X, or $\lim_{x\to x_0} f(x) = f(x_0)$. f is **continuous** if it is continuous at every point.

Proposition 3.1 (3.1.1)

If $\lim_{x\to x_0} f(x) = y_0$ and $\lim_{x\to x_0} g(x) = z_0$, then $f + g \to y_0 + z_0$, $af \to ay_0$, and for real-valued functions $fg \to y_0 z_0$, and $f/g \to y_0/z_0$ if $z_0 \neq 0$.

Proof. As in Theorem 1.14, or by Proposition 3.3 below applied to sequences. ■

Proposition 3.2 (3.1.2)

If $f, g : X \to \mathbb{R}^m$ are continuous, so is $af + bg$; for real-valued f, g, so are fg and f/g (where $g \neq 0$).

Proof. Apply Proposition 3.1 at each non-isolated point. ■

Polynomials in several variables are therefore continuous, and so are rational functions where the denominator does not vanish.

Proposition 3.3 (3.1.3, sequential criterion)

f is continuous at x_0 iff $f(x_k) \to f(x_0)$ for every sequence $x_k \to x_0$ in X.

Proof. ($\Rightarrow$) Given ϵ, take δ; for large k, $\|x_k - x_0\| < \delta$, so $\|f(x_k) - f(x_0)\| < \epsilon$.

($\Leftarrow$) If f is not continuous at x_0, for some ϵ and each k there is x_k with $\|x_k - x_0\| < 1/k$ but $\|f(x_k) - f(x_0)\| \geq \epsilon$. ■

Similarly $\lim_{x\to x_0} f(x) = y_0$ iff $f(x_k) \to y_0$ for every sequence $x_k \to x_0$ with $x_k \neq x_0$.

Proposition 3.4 (3.1.4)

If $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.

Proof. Given ϵ, choose η for g at $f(x_0)$ and then δ for f at x_0 with η in place of ϵ. ■

Example 3.1

(a) $f(x) = \sin(1/x)$ $(x \neq 0)$ has no limit at 0: $x_k = 1/(k\pi + \pi/2)$ gives ± 1 alternately.
(b) $x \sin(1/x) \to 0$ by the squeeze $|x \sin(1/x)| \leq |x|$.

(c) Thomae's function ($f(p/q) = 1/q$ in lowest terms, $f = 0$ at irrationals) is continuous exactly at the irrationals.
(d) $f(x,y) = xy/(x^2+y^2)$, $f(0,0) = 0$, is continuous along every line through the origin, but $f(t,t) = \frac{1}{2}$, so f is not continuous at 0.

Theorem 3.5 (3.1.5)

For $f : X \to Y$ the following are equivalent:
(a) f is continuous;
(b) $f^{-1}(V)$ is open in X for every open V in Y;
(c) $f^{-1}(E)$ is closed in X for every closed E in Y.

Proof. (a)⇒(b): if $x \in f^{-1}(V)$ and $N(f(x),\epsilon) \cap Y \subset V$, the δ of continuity gives $N(x,\delta) \cap X \subset f^{-1}(V)$. (b)⇒(a): $f^{-1}(N(f(x_0),\epsilon) \cap Y)$ is open in X and contains x_0. (b)⇔(c): $f^{-1}(Y - E) = X - f^{-1}(E)$. ■

3.2 Continuity, compactness and connectedness

Motivation. Extreme and intermediate values express that continuity preserves compactness and connectedness, in any $\mathbb{R}^n$.

Theorem 3.6 (3.2.1)

If $f : X \to Y$ is continuous and surjective and X is compact, then Y is compact.

Proof. Let $\{U_i\}$ be an open cover of Y; by Theorem 3.5 and Proposition 2.6, $f^{-1}(U_i) = V_i \cap X$ with V_i open in $\mathbb{R}^n$, and the V_i cover X. Finitely many $V_{i_1}, \ldots, V_{i_k}$ do; then $U_{i_1}, \ldots, U_{i_k}$ cover $Y = f(X)$. ■

Corollary 3.7 (3.2.2, extreme value theorem)

A continuous $f : X \to \mathbb{R}$ on a compact set attains its maximum and minimum.

Proof. $f(X)$ is compact, hence bounded and closed; so $\sup f(X)$ is finite and, being a limit of points of $f(X)$, lies in $f(X)$. ■

Corollary 3.8 (3.2.3)

If X is compact and $f : X \to \mathbb{R}$ is continuous with $f > 0$, then there is $\delta > 0$ with $f(x) \geq \delta$ for all x.

Proof. Take $\delta = \min f > 0$. ■

Corollary 3.9 (3.2.4)

A continuous $f : [a,b] \to \mathbb{R}$ attains its maximum and minimum.

Corollary 3.10 (3.2.5)

If X is compact and $f : X \to Y$ is a continuous bijection, then f^{-1} is continuous.

Proof. Let U be open in X. $X - U$ is closed in the compact X, so compact (Proposition 2.24); then $f(X - U) = Y - f(U)$ is compact, hence closed, so $f(U)$ is open in Y. By Theorem 3.5, f^{-1} is continuous. ■

Compactness is needed: $t \mapsto (\cos t, \sin t)$ is a continuous bijection $[0, 2\pi) \to S^1$ whose inverse is not continuous. For nonempty $A, B \subset \mathbb{R}^n$ let $\operatorname{dist}(A, B) = \inf\{\|a - b\| : a \in A, b \in B\}$.

Proposition 3.11 (3.2.6)

If A is compact, B is closed and $A \cap B = \emptyset$, then $\operatorname{dist}(A, B) > 0$.

Proof. The function $d(x) = \operatorname{dist}(\{x\}, B)$ is continuous (indeed $|d(x) - d(y)| \le \|x - y\|$) and positive on A, since $\mathbb{R}^n - B$ is open. By Corollary 3.8 it is bounded below on A by some $\delta > 0$. ■

The compactness is needed: the closed sets $\{(x, 0)\}$ and $\{(x, 1/x) : x > 0\}$ are at distance 0.

Lemma 3.12 (3.2.7)

S is disconnected iff there is a continuous surjection $f : S \to \{0, 1\}$.

Proof. If U, V disconnect S, let $f = 0$ on $U \cap S$ and 1 on $V \cap S$; preimages of subsets of $\{0, 1\}$ are relatively open, so f is continuous. Conversely, $f^{-1}(0)$ and $f^{-1}(1)$ are nonempty and relatively open (Theorem 3.5), so they are $U \cap S$ and $V \cap S$ with U, V open (Proposition 2.6); then $U \cap V \cap S = f^{-1}(0) \cap f^{-1}(1) = \emptyset$ and $S \subset U \cup V$. ■

Theorem 3.13 (3.2.8)

If $f : X \to Y$ is continuous and surjective and X is connected, then Y is connected.

Proof. If Y were disconnected, Lemma 3.12 gives a continuous surjection $g : Y \to \{0, 1\}$, and then $g \circ f : X \to \{0, 1\}$ is a continuous surjection. ■

Corollary 3.14 (3.2.9)

If $I \subset \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$ is continuous, then $f(I)$ is an interval.

Proof. Proposition 2.28 and Theorem 3.13. ■

Corollary 3.15 (3.2.10, intermediate value theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < \alpha < f(b)$, then $f(c) = \alpha$ for some $c \in (a, b)$.

Corollary 3.16 (3.2.11)

If $f : [a, b] \to \mathbb{R}$ is continuous, then $f([a, b])$ is a bounded closed interval $[m, M]$.

Proof. It is an interval (Corollary 3.14) and compact (Theorem 3.6), containing its minimum m and maximum M. ■

A function mapping intervals to intervals need not be continuous, for example $\sin(1/x)$ extended by 0 at 0.

Corollary 3.17 (3.2.12, fixed point)

Every continuous $f : [a, b] \to [a, b]$ has a fixed point.

Proof. $g(x) = x - f(x)$ has $g(a) \le 0 \le g(b)$; apply Corollary 3.15 (or $g(a) = 0$ or $g(b) = 0$). ■

Example 3.2

Every polynomial of odd degree has a real root (it takes both signs). Every $a > 0$ has a unique positive n-th root, since x^n is continuous, strictly increasing and unbounded on $[0, \infty)$.

3.3 Uniform continuity

Motivation. One δ for all points is what integration needs; on compact sets continuity gives it.

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(y)\| < \epsilon$ whenever $x, y \in X$, $\|x - y\| < \delta$.

Theorem 3.18 (3.3.1, Heine)

A continuous function on a compact set is uniformly continuous.

Proof. Given ϵ, choose for each $x \in X$ a $\delta(x) > 0$ with $\|f(y)-f(x)\| < \epsilon/2$ for $y \in X$, $\|y-x\| < \delta(x)$. The sets $N(x, \delta(x)/2)$ cover X; finitely many with centres $x_1, \dots, x_k$ do. Let $\delta = \min \delta(x_i)/2$. If $\|x - y\| < \delta$, then $\|x - x_i\| < \delta(x_i)/2$ for some i, and $\|y - x_i\| < \delta(x_i)$; so $\|f(x) - f(y)\| \le \|f(x) - f(x_i)\| + \|f(x_i) - f(y)\| < \epsilon$. ■

Example 3.3

x^2 is not uniformly continuous on $\mathbb{R}$ ($(n+1/n)^2 - n^2 > 2$), and $1/x$ is not uniformly continuous on $(0, 1)$; $\sqrt{x}$ is uniformly continuous on $[0, \infty)$ since $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$.

Theorem 3.19 (3.3.2)

A uniformly continuous function maps Cauchy sequences to Cauchy sequences.

Proof. Given ϵ, take δ from uniform continuity and N with $\|x_m - x_n\| < \delta$ for $m, n \ge N$. ■

Corollary 3.20 (3.3.3)

A uniformly continuous $f : X \to \mathbb{R}^m$ extends uniquely to a continuous function on $\overline{X}$, which is again uniformly continuous.

Proof. For $x_0 \in \overline{X}$ and $X \ni x_k \to x_0$, $(f(x_k))$ is Cauchy (Theorem 3.19), hence convergent (Theorem 2.14); let $F(x_0)$ be the limit, independent of the sequence (interlace two), with $F = f$ on X. If $\|x - y\| < \delta$ in $\overline{X}$, take $x_k \to x$, $y_k \to y$ with $\|x_k - y_k\| < \delta$: $\|F(x) - F(y)\| = \lim \|f(x_k) - f(y_k)\| \le \epsilon$. Uniqueness follows from Proposition 3.3. ■

3.4 Monotone functions

Motivation. Monotone functions have only jump discontinuities, and countably many.

For $f : (a, b) \to \mathbb{R}$ write $f(x+) = \lim_{t \to x^+} f(t)$ and $f(x-) = \lim_{t \to x^-} f(t)$ when these exist.

Theorem 3.21 (3.4.1)

A monotone $f : (a, b) \to \mathbb{R}$ has one-sided limits at every point, and $f(a+)$, $f(b-)$ exist in $[-\infty, \infty]$. If f is increasing and $a < x < y < b$, then $f(x-) \le f(x) \le f(x+) \le f(y-) \le f(y) \le f(y+)$, and $f(x+) = \inf_{t>x} f(t)$, $f(x-) = \sup_{t<x} f(t)$.

Proof. Let f increase. $A = \{f(t) : a < t < x\}$ is bounded above by $f(x)$; let $s = \sup A$. Given ϵ, some $f(t_0) > s - \epsilon$, and for $t_0 < t < x$, $s - \epsilon < f(t_0) \le f(t) \le s$; so $f(x-) = s \le f(x)$. Similarly $f(x+) = \inf_{t>x} f(t) \ge f(x)$. For $x < y$, choosing $x < t < y$, $f(x+) \le f(t) \le f(y-)$. ■

Corollary 3.22 (3.4.2)

The set of discontinuities of a monotone function on an interval is countable.

Proof. For increasing f, f is discontinuous at x iff $f(x-) < f(x+)$. Choose a rational $r(x) \in (f(x-), f(x+))$; by Theorem 3.21 these open intervals are disjoint for distinct x, so $x \mapsto r(x)$ is injective into $\mathbb{Q}$. ■

Conversely, for any countable set $\{x_k\} \subset (a, b)$, $f(x) = \sum_{x_k < x} 2^{-k}$ is increasing and discontinuous exactly at the x_k.

Theorem 3.23 (3.4.3)

For a continuous $f : I \to \mathbb{R}$ on an interval, f is injective iff it is strictly monotone. In that case $f(I)$ is an interval and $f^{-1} : f(I) \to I$ is continuous.

Proof. Strictly monotone implies injective. Conversely, if f is injective but not strictly monotone, some $x < y < z$ in I have, say, $f(y) > \max(f(x), f(z))$; any $\alpha \in (\max(f(x), f(z)), f(y))$ is attained in (x, y) and in (y, z) (Corollary 3.15), a contradiction. For strictly increasing f, f^{-1} is strictly increasing on the interval $f(I)$ (Corollary 3.14) with image the interval I; a jump $g(y-) < g(y+)$ would omit values (Theorem 3.21), so f^{-1} is continuous. ■

3.5 Selected exercises

Motivation. Four book exercises using density, compactness, connectedness and completeness.

$F \subset \mathbb{R}^n$ is closed iff it contains the limits of its convergent sequences (Proposition 2.3(b), Theorem 2.4).

Exercise 3.5.2 (Cauchy's functional equation)

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with $f(x + y) = f(x) + f(y)$ for all x, y. Show that $f(x) = ax$ for all x, where $a = f(1)$.

Solution.

Step 1. $f(0) = f(0 + 0) = 2f(0)$, so $f(0) = 0$; then $0 = f(r + (-r)) = f(r) + f(-r)$ gives $f(-r) = -f(r)$.

Step 2. $f((n+1)r) = f(nr) + f(r)$ gives $f(nr) = nf(r)$ for $n \in \mathbb{N}$ by induction, and for $n \in \mathbb{Z}$ by Step 1.

Step 3. $a = f(n \cdot \frac{1}{n}) = nf(\frac{1}{n})$, so $f(\frac{m}{n}) = mf(\frac{1}{n}) = a\frac{m}{n}$: $f(q) = aq$ on $\mathbb{Q}$.

Step 4. For $x \in \mathbb{R}$ take $q_k \in \mathbb{Q} \cap (x, x+1/k)$ (Theorem 1.8); by Proposition 3.3, $f(x) = \lim aq_k = ax$. ■

Exercise 3.5.4 (Closed graphs)

For $f : \mathbb{R} \to \mathbb{R}$ let $G = \{(x, f(x)) : x \in \mathbb{R}\} \subset \mathbb{R}^2$ be its graph.
(a) Show that if f is continuous, then G is closed.
(b) Show that if f is bounded and G is closed, then f is continuous.
(c) Show by an example that boundedness cannot be dropped in (b).

Solution.

(a) If $(x_k, f(x_k)) \to (x, y)$, then $x_k \to x$, so $y = \lim f(x_k) = f(x)$ (Proposition 3.3) and $(x, y) \in G$.

(b)

Step 1. Let $|f| \le M$ and f be discontinuous at x: there are $\epsilon > 0$, $x_k \to x$ with $|f(x_k) - f(x)| \ge \epsilon$ (Proposition 3.3).

Step 2. A subsequence $f(x_{k_j}) \to y$ (Theorem 2.10), with $|y - f(x)| \ge \epsilon$.

Step 3. $(x_{k_j}, f(x_{k_j})) \to (x, y) \in G$, so $y = f(x)$, contradicting Step 2.

(c) $f(x) = 1/x$ $(x \neq 0)$, $f(0) = 0$ is discontinuous at 0 with closed graph: let $(x_k, f(x_k)) \to (x, y)$.

Step 1. $x \neq 0$: $f(x_k) = 1/x_k \to 1/x$ for large k, so $y = f(x)$.

Step 2. $x = 0$: $|f(x_k)| \le K$, so $x_k \neq 0$ forces $|x_k| \ge 1/K$; as $x_k \to 0$, $x_k = 0$ eventually and $y = 0 = f(0)$. ■

Exercise 3.5.9 (Antipodal points with equal values)

Let $T = \{(\cos t, \sin t) : t \in \mathbb{R}\} \subset \mathbb{R}^2$ be the unit circle and $f : T \to \mathbb{R}$ continuous. Show that $f(x) = f(-x)$ for some $x \in T$.

Solution.

Step 1. T is the continuous image of $[0, 2\pi]$, hence connected (Proposition 2.28, Theorem 3.13).

Step 2. $g(x) = f(x) - f(-x)$ is continuous on T (Propositions 3.2, 3.4), so $g(T)$ is an interval (Theorem 3.13, Proposition 2.28).

Step 3. For $p = (1, 0)$, $g(-p) = -g(p)$, so $0 \in g(T)$: some x has $f(x) = f(-x)$. ■

Exercise 3.5.10 (Contraction mapping principle)

Let $X \subset \mathbb{R}^n$ be nonempty and closed, $0 < r < 1$, and $f : X \to X$ with $\|f(x) - f(y)\| \le r\|x - y\|$ for all $x, y \in X$.
(a) Show that f is uniformly continuous.
(b) For $x_0 \in X$ let $x_{k+1} = f(x_k)$. Show that (x_k) is a Cauchy sequence.
(c) Show that f has exactly one fixed point x, that $x_k \to x$, and that $\|x_k - x\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Solution.

(a) $\delta = \epsilon$: $\|x - y\| < \delta \Rightarrow \|f(x) - f(y)\| \le r\|x - y\| < \epsilon$.

(b)

Step 1. By induction, $\|x_{j+1} - x_j\| = \|f(x_j) - f(x_{j-1})\| \le r\|x_j - x_{j-1}\| \le \cdots \le r^j\|x_1 - x_0\|$.

Step 2. For $m > k$, by the triangle inequality,

$$\|x_m - x_k\| \le \sum_{j=k}^{m-1} \|x_{j+1} - x_j\| \le \|x_1 - x_0\| \sum_{j=k}^{\infty} r^j = \frac{r^k}{1-r}\|x_1 - x_0\|.$$

Step 3. $r^k \to 0$ (Example 1.1(b)), so (x_k) is Cauchy.

(c)

Step 1. $x_k \to x \in \mathbb{R}^n$ (Theorem 2.14), and $x \in X$ as X is closed.

Step 2. By (a), $f(x) = \lim f(x_k) = \lim x_{k+1} = x$ (Proposition 3.3).

Step 3. $f(y) = y$ gives $\|x - y\| \le r\|x - y\|$, so $y = x$.

Step 4. $m \to \infty$ in Step 2 of (b) gives $\|x - x_k\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Closedness is needed: $f(x) = x/2$ on $X = (0, 1]$ is a contraction without a fixed point. ■

Chapter 4. Differentiable Functions

4.1 Differentiability

Motivation. The derivative is the best linear approximation; its rules follow from $f(x+h) = f(x) + f'(x)h + o(h)$.

Let I be an interval. $f : I \to \mathbb{R}$ is **differentiable** at $x_0 \in I$ if $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists (one-sided at an end point).

Proposition 4.1 (4.1.1)

For $f : (a, b) \to \mathbb{R}$ and $x_0 \in (a, b)$ the following are equivalent:
(a) f is differentiable at x_0;
(b) there are $\lambda \in \mathbb{R}$ and a function ϕ with $f(x_0+h)-f(x_0) = \lambda h+|h|\phi(h)$ and $\lim_{h\to 0} \phi(h) = 0$.
Then $\lambda = f'(x_0)$.

Proof. (a)⇒(b): put $\lambda = f'(x_0)$ and $\phi(h) = (f(x_0 + h) - f(x_0) - \lambda h)/|h|$ for $h \neq 0$, $\phi(0) = 0$. (b)⇒(a): $\frac{f(x_0+h)-f(x_0)}{h} = \lambda \pm \phi(h) \to \lambda$. ■

This linear-approximation form generalizes to several variables.

Proposition 4.2 (4.1.2)

If f is differentiable at x_0, it is continuous at x_0.

Proof. $f(x_0 + h) - f(x_0) = \lambda h + |h|\phi(h) \to 0$. ■

Proposition 4.3 (4.1.3)

If f, g are differentiable at x_0, so are $f+g$, fg, and f/g (if $g(x_0) \neq 0$), with $(f+g)' = f'+g'$, $(fg)' = f'g + fg'$, $(f/g)' = (f'g - fg')/g^2$ at x_0.

Proof. For the product, $\frac{f(x_0+h)g(x_0+h)-f(x_0)g(x_0)}{h} = \frac{f(x_0+h)-f(x_0)}{h} g(x_0+h) + f(x_0)\frac{g(x_0+h)-g(x_0)}{h}$, and $g(x_0 + h) \to g(x_0)$ by Proposition 4.2. For the quotient, first $\frac{1/g(x_0+h)-1/g(x_0)}{h} = -\frac{g(x_0+h)-g(x_0)}{h} \cdot \frac{1}{g(x_0+h)g(x_0)} \to -g'/g^2$, then use the product rule. ■

Proposition 4.4 (4.1.4, chain rule)

If $f : I \to J$ is differentiable at x_0 and $g : J \to \mathbb{R}$ is differentiable at $y_0 = f(x_0)$, then $g \circ f$ is differentiable at x_0 and $(g \circ f)'(x_0) = g'(y_0)f'(x_0)$.

Proof. Write $g(y_0 + k) - g(y_0) = k(g'(y_0) + \psi(k))$ with $\psi(k) \to 0 = \psi(0)$. With $k = f(x_0 + h) - f(x_0) \to 0$,

$$\frac{g(f(x_0 + h)) - g(f(x_0))}{h} = \frac{k}{h}\,(g'(y_0) + \psi(k)) \to f'(x_0)g'(y_0).$$

■

Example 4.1

$f(x) = x^2 \sin(1/x)$, $f(0) = 0$, is differentiable everywhere with $f'(0) = \lim h \sin(1/h) = 0$, but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ is not continuous at 0. $|x|$ is continuous but not differentiable at 0.

4.2 Mean value theorems

Motivation. The mean value theorem turns local derivative information into global conclusions.

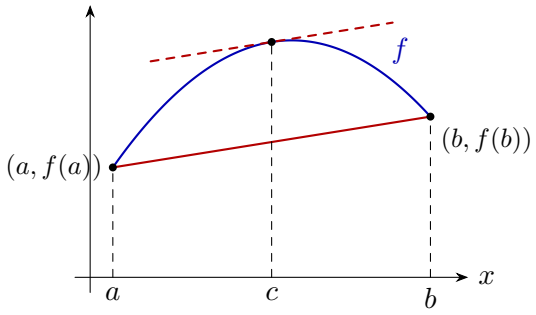


Figure. *Mean value theorem: the tangent at c (dashed) is parallel to the secant.*

If f has a local maximum or minimum at an interior point c and is differentiable there, then $f'(c) = 0$: the difference quotients have opposite signs on the two sides.

Lemma 4.5 (4.2.1, Rolle)

If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

Proof. If f is constant, any c works. Otherwise f attains a maximum or minimum (Corollary 3.9) at a point $c \in (a, b)$ where $f(c) \neq f(a)$, and $f'(c) = 0$. ∎

Theorem 4.6 (4.2.2, Cauchy mean value theorem)

If $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b), there is $c \in (a, b)$ with $f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]$.

Proof. Apply Rolle to $h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)]$, for which $h(a) = h(b) = f(a)g(b) - g(a)f(b)$. ∎

Corollary 4.7 (4.2.3, mean value theorem)

If f is continuous on $[a, b]$ and differentiable on (a, b), then $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$.

Proof. Take $g(x) = x$. ∎

Corollary 4.8 (4.2.4)

For differentiable $f : (a, b) \to \mathbb{R}$: f is increasing iff $f' \geq 0$; constant iff $f' = 0$; decreasing iff $f' \leq 0$. If $f' > 0$, f is strictly increasing.

Proof. If $f' \geq 0$ and $x < y$, $f(y) - f(x) = f'(c)(y - x) \geq 0$. Conversely, if f increases, its difference quotients are ≥ 0, so $f' \geq 0$. ■

Theorem 4.9 (4.2.5, L'Hôpital's rule)

Let f, g be differentiable near x_0 (except possibly at x_0) with $g' \neq 0$ there, $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = 0$, and $\lim_{x\to x_0} f'(x)/g'(x) = L$. Then $\lim_{x\to x_0} f(x)/g(x) = L$. The same holds for one-sided limits, for $x \to \pm\infty$, and when $g(x) \to \pm\infty$.

Proof. Set $f(x_0) = g(x_0) = 0$. Near x_0, $g(x) \neq 0$ (Rolle, $g' \neq 0$), and Theorem 4.6 gives $f(x)/g(x) = f'(c_x)/g'(c_x)$ with c_x between x_0 and x, so $c_x \to x_0$. For $x \to \infty$ put $x = 1/t$.

If $g \to \infty$ as $x \to x_0^+$: choose δ with $|f'/g' - L| < \epsilon$ on $(x_0, x_0 + \delta)$, fix y there; for $x_0 < x < y$, $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(c)}{g'(c)}$ and

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(y)}{g(x) - g(y)} \cdot \frac{1 - g(y)/g(x)}{1} + \frac{f(y)}{g(x)},$$

where $g(y)/g(x) \to 0$ and $f(y)/g(x) \to 0$ as $x \to x_0^+$; so $\limsup |f/g - L| \leq \epsilon$. ■

Theorem 4.10 (4.2.6, Darboux)

If $F : [a, b] \to \mathbb{R}$ is differentiable and $F'(a) < \alpha < F'(b)$, then $F'(c) = \alpha$ for some $c \in (a, b)$.

Proof. $G(x) = F(x) - \alpha x$ has $G'(a) < 0 < G'(b)$, so $G(x) < G(a)$ for x slightly larger than a and $G(x) < G(b)$ for x slightly smaller than b. Hence the minimum of G on $[a, b]$ is attained at an interior point c, where $G'(c) = 0$. ■

Corollary 4.11 (4.2.7)

If $f : I \to \mathbb{R}$ has a primitive on the interval I, then $f(J)$ is an interval for every interval $J \subset I$.

Proof. Apply Theorem 4.10 to the primitive on subintervals of J. ■

So a function with a jump discontinuity, such as the sign function, has no primitive; derivatives can be discontinuous (Example 4.1) but have the intermediate value property.

4.3 Taylor's theorem

Motivation. Taylor's theorem bounds the error of approximating f by its Taylor polynomial.

For f with n derivatives at a, let $T_n^f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k$ be its Taylor polynomial at a.

Theorem 4.12 (4.3.1, generalized mean value theorem)

Let I be the closed interval with end points $a \neq b$, and $f, g : I \to \mathbb{R}$ with $f^{(n)}, g^{(n)}$ continuous on I and differentiable on $\operatorname{Int} I$. For each $x \in I$, $x \neq a$, there is c strictly between a and x

with

$$[f(x)-T_n^f(x)]g^{(n+1)}(c)=[g(x)-T_n^g(x)]f^{(n+1)}(c).$$

Proof. Fix x and let $F(t)=\sum_{k=0}^{n}\frac{f^{(k)}(t)}{k!}(x-t)^k$, $G(t)$ likewise for g, for t between a and x. In $F'(t)$ the terms telescope:

$$F'(t)=\sum_{k=0}^{n}\frac{f^{(k+1)}(t)}{k!}(x-t)^k-\sum_{k=1}^{n}\frac{f^{(k)}(t)}{(k-1)!}(x-t)^{k-1}=\frac{f^{(n+1)}(t)}{n!}(x-t)^n,$$

and similarly for G. Also $F(x)=f(x)$ and $F(a)=T_n^f(x)$. Theorem 4.6 gives c with $[F(x)-F(a)]G'(c)=[G(x)-G(a)]F'(c)$; cancel the nonzero factor $(x-c)^n/n!$. ■

Corollary 4.13 (4.3.2, Taylor's theorem)

Under the hypotheses on f, for each $x\in I$ there is c between a and x with

$$f(x)=\sum_{k=0}^{n}\frac{f^{(k)}(a)}{k!}(x-a)^k+\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Apply Theorem 4.12 with $g(t)=(t-a)^{n+1}$: $T_n^g=0$ and $g^{(n+1)}=(n+1)!$. ■

The last term is the **remainder**. If f is C^∞, the **Taylor series** of f at a is $\sum_k\frac{f^{(k)}(a)}{k!}(x-a)^k$; it may fail to converge to f: $f(x)=e^{-1/x^2}$, $f(0)=0$, has all derivatives 0 at 0.

Proposition 4.14 (4.3.3)

Let f be C^∞ on I, $a,x\in I$, and J the closed interval between them. If $\{|f^{(n)}(t)|:t\in J,n\ge 0\}$ is bounded, the Taylor series of f at a converges to $f(x)$.

Proof. With M a bound, the remainder is at most $M|x-a|^{n+1}/(n+1)!\to 0$ (the terms of the convergent series $\sum r^n/n!$ tend to 0). ■

Example 4.2

e^x, $\sin x$, $\cos x$ satisfy the hypothesis on every bounded interval, so $e^x=\sum x^n/n!$, $\sin x=\sum(-1)^n x^{2n+1}/(2n+1)!$, $\cos x=\sum(-1)^n x^{2n}/(2n)!$ for all real x. Taylor's theorem also gives the second derivative test: if $f'(a)=0$ and $f''(a)>0$, f has a strict local minimum at a.

4.4 Selected exercises

Motivation. Four book exercises using the mean value theorem, Rolle and L'Hôpital.

Exercise 4.4.8 (A Hölder condition of order >1)

Let I be an interval, $M>0$, $\alpha>1$, and $f:I\to\mathbb{R}$ with $|f(x)-f(y)|\le M|x-y|^\alpha$ for all $x,y\in I$. Show that f is constant. (The book's case is $M=1$, $\alpha=2$, $I=\mathbb{R}$.)

Solution.

Step 1. For $x \in I$ and $y \in I$, $y \neq x$,

$$\Big|\frac{f(y)-f(x)}{y-x}\Big| \leq M|y-x|^{\alpha-1} \longrightarrow 0 \quad (y \to x),$$

since $\alpha - 1 > 0$. So f is differentiable on I with $f' \equiv 0$ (one-sided at end points).

Step 2. For $a < b$ in I, $f(b) - f(a) = f'(c)(b-a) = 0$ (Proposition 4.2, Corollary 4.7).

It fails for $\alpha = 1$ ($f(x) = x$) and $\alpha = \frac{1}{2}$ ($\sqrt{x}$ on $[0,\infty)$, Example 3.3). ■

Exercise 4.4.9 (Limits of the derivative)

Let I be an open interval, $c \in I$, and $f : I \to \mathbb{R}$ continuous at c and differentiable on $I - \{c\}$. Show that if $\lim_{x\to c} f'(x) = K$ exists, then f is differentiable at c and $f'(c) = K$.

Solution.

Step 1. Let $\epsilon > 0$. Choose $\eta > 0$ with $[c-\eta, c+\eta] \subset I$ and $|f'(z) - K| < \epsilon$ for $0 < |z-c| < \eta$.

Step 2. For $0 < h < \eta$, f is continuous on $[c, c+h]$ and differentiable inside, so by Corollary 4.7 some $z \in (c, c+h)$ has

$$\frac{f(c+h)-f(c)}{h} = f'(z), \qquad \text{so} \qquad \Big|\frac{f(c+h)-f(c)}{h} - K\Big| < \epsilon.$$

Step 3. Likewise for $-\eta < h < 0$; so $f'(c) = K$.

Hence derivatives have no removable or jump discontinuities; a discontinuous derivative (Example 4.1) has no limit there. ■

Exercise 4.4.12 (Roots of $\frac{d^n}{dx^n}(x^2-x)^n$)

Let $n \geq 1$ and $f(x) = (x^2-x)^n = x^n(x-1)^n$. Show that $f^{(n)}$ has exactly n distinct roots, all in $(0,1)$.

Solution.

Step 1. For $0 \leq k \leq n$, $f^{(k)}(x) = x^{n-k}h_k(x)$ for some polynomial h_k. This holds for $k = 0$ with $h_0 = (x-1)^n$; if it holds for $k < n$, then with $m = n - k \geq 1$,

$$f^{(k+1)}(x) = mx^{m-1}h_k(x) + x^m h_k'(x) = x^{m-1}(mh_k(x) + xh_k'(x)).$$

Hence $f^{(k)}(0) = 0$ for $0 \leq k \leq n-1$. Writing $f(x) = (x-1)^n x^n$, the same argument with $x-1$ in place of x gives $f^{(k)}(1) = 0$ for $0 \leq k \leq n-1$.

Step 2. Claim: $f^{(k)}$ $(1 \leq k \leq n)$ has $\geq k$ distinct roots in $(0,1)$. $k = 1$: Rolle (Lemma 4.5) on $[0,1]$. If $k \leq n-1$ and $0 < \alpha_1 < \cdots < \alpha_k < 1$ are roots of $f^{(k)}$, by Step 1 $f^{(k)}$ vanishes at the $k+2$ points

$$0 < \alpha_1 < \cdots < \alpha_k < 1.$$

and Rolle on the $k+1$ disjoint gaps gives $k+1$ distinct roots of $f^{(k+1)}$.

Step 3. $f^{(n)}$ has degree n (leading coefficient $(2n)!/n!$), hence at most n roots; with Step 2, exactly n, all in $(0,1)$.

Up to a constant and $x = (1+t)/2$, $f^{(n)}$ is the n-th Legendre polynomial. ■

Exercise 4.4.13 (Symmetric second difference quotient)

(a) Let f be differentiable on an open interval $I \ni c$ and suppose $f''(c)$ exists. Show that

$$\lim_{h\to 0} \frac{f(c+h)+f(c-h)-2f(c)}{h^2} = f''(c).$$

(b) Show that the limit in (a) may exist although $f''(c)$ does not.

Solution.

(a)

Step 1. $N(h) = f(c+h)+f(c-h)-2f(c)$ and $D(h) = h^2$ are differentiable, tend to 0, and $D'(h) = 2h \neq 0$.

Step 2.

$$\frac{N'(h)}{D'(h)} = \frac{f'(c+h)-f'(c-h)}{2h} = \frac{1}{2}\Big(\frac{f'(c+h)-f'(c)}{h} + \frac{f'(c-h)-f'(c)}{-h}\Big)$$
$$\longrightarrow \frac{1}{2}(f''(c)+f''(c)) = f''(c).$$

Step 3. By L'Hôpital's rule (Theorem 4.9), $N(h)/D(h) \to f''(c)$.

(b) $f(x) = x|x|$, $c = 0$: $N(h) = 0$, so the limit is 0; but $f'(x) = 2|x|$ (at 0: $f(h)/h = |h| \to 0$) is not differentiable at 0. ■

Chapter 5. The Riemann–Stieltjes Integral

5.1 The Riemann integral

Motivation. The Riemann integral squeezes the area under a graph between upper and lower sums.

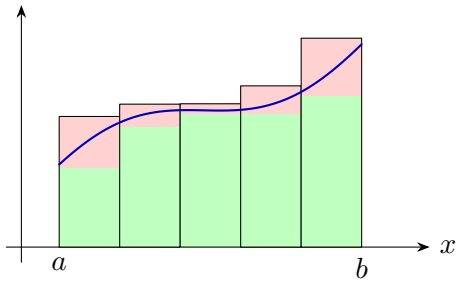


Figure. *Lower sum (green) and upper sum (green plus red); f is integrable when refinement makes the red excess $U(f,P)-L(f,P)$ arbitrarily small.*

A **partition** of $[a,b]$ is a finite set $P=\{a=x_0<x_1<\cdots<x_n=b\}$; write $\Delta x_i=x_i-x_{i-1}$ and $\|P\|=\max\Delta x_i$ (the mesh). $Q\supset P$ is a **refinement** of P. For bounded $f:[a,b]\to\mathbb{R}$ let $M_i=\sup_{[x_{i-1},x_i]}f$, $m_i=\inf_{[x_{i-1},x_i]}f$, and

$$U(f,P)=\sum M_i\Delta x_i,\qquad L(f,P)=\sum m_i\Delta x_i.$$

The **upper** and **lower integrals** are $\overline{\int}f=\inf_P U(f,P)$ and $\underline{\int}f=\sup_P L(f,P)$; f is **Riemann integrable** ($f\in\mathcal{R}$) if they are equal, and the common value is $\int_a^b f$.

Proposition 5.1 (5.1.1)

If $P\subset Q$, then $L(f,P)\le L(f,Q)\le U(f,Q)\le U(f,P)$. For any partitions P,P', $L(f,P)\le U(f,P')$; hence $\underline{\int}f\le\overline{\int}f$.

Proof. It suffices to add one point $y\in(x_{i-1},x_i)$: the infima of f over $[x_{i-1},y]$ and $[y,x_i]$ are $\ge m_i$, so L can only increase; similarly U decreases. Then $L(f,P)\le L(f,P\cup P')\le U(f,P\cup P')\le U(f,P')$. ∎

Proposition 5.2 (5.1.2, Riemann's criterion)

A bounded f is integrable iff for every $\epsilon>0$ there is a partition P with $U(f,P)-L(f,P)<\epsilon$.

Proof. If the condition holds, $0\le\overline{\int}f-\underline{\int}f\le U(f,P)-L(f,P)<\epsilon$. Conversely, if $f\in\mathcal{R}$, choose P_1,P_2 with $U(f,P_1)<\int f+\epsilon/2$ and $L(f,P_2)>\int f-\epsilon/2$; then $P=P_1\cup P_2$ works by Proposition 5.1. ∎

Theorem 5.3 (5.1.3)

If $f,g\in\mathcal{R}$ and $c\in\mathbb{R}$, then $f+g, cf\in\mathcal{R}$ with $\int(f+g)=\int f+\int g$ and $\int cf=c\int f$; and $f\le g$ implies $\int f\le\int g$.

Proof. On each subinterval $\sup(f+g)\le\sup f+\sup g$ and $\inf(f+g)\ge\inf f+\inf g$, so

$$L(f,P)+L(g,P)\le L(f+g,P)\le U(f+g,P)\le U(f,P)+U(g,P).$$

Choosing P with $U - L < \epsilon/2$ for both f and g (common refinement) gives integrability and the formula. For $c \geq 0$, $U(cf, P) = cU(f, P)$ and $L(cf, P) = cL(f, P)$; for $c = -1$, $U(-f, P) = -L(f, P)$. If $f \leq g$, then $U(f, P) \leq U(g, P)$ for all P. ■

Products of integrable functions are integrable: $f^2 \in \mathcal{R}$ because on each subinterval $\sup f^2 - \inf f^2 \leq 2K(M_i - m_i)$ with $K = \sup|f|$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

Theorem 5.4 (5.1.4)

If $f \in \mathcal{R}[a, b]$ and $a < c < b$, then $f \in \mathcal{R}[a, c]$, $f \in \mathcal{R}[c, b]$ and $\int_a^b f = \int_a^c f + \int_c^b f$. Conversely, integrability on $[a, c]$ and $[c, b]$ implies integrability on $[a, b]$.

Proof. For partitions P of $[a, b]$ containing c, with $P_1 = P \cap [a, c]$ and $P_2 = P \cap [c, b]$, we have $U(f, P) = U(f, P_1) + U(f, P_2)$, likewise for L. If $U(f, P) - L(f, P) < \epsilon$ (we may add c to P), the same holds for P_1 and P_2. Conversely, combine partitions of the two halves. Taking infima of upper sums gives the formula. ■

With the convention $\int_b^a f = -\int_a^b f$, the formula holds for any order of a, b, c.

Theorem 5.5 (5.1.5)

If $f \in \mathcal{R}$, then $|f| \in \mathcal{R}$ and $|\int_a^b f| \leq \int_a^b |f|$.

Proof. From $||f(x)| - |f(y)|| \leq |f(x) - f(y)|$, the oscillation of $|f|$ on each subinterval is at most that of f, so $U(|f|, P) - L(|f|, P) \leq U(f, P) - L(f, P)$. Then $-|f| \leq f \leq |f|$ and Theorem 5.3 give the inequality. ■

Example 5.1

The Dirichlet function (1 on $\mathbb{Q}$, 0 elsewhere) is not integrable on $[0, 1]$: every $U(f, P) = 1$ and every $L(f, P) = 0$. Integrability of $|f|$ does not imply that of f: take $f = 1$ on $\mathbb{Q}$ and -1 elsewhere.

5.2 Integrable functions

Motivation. Continuous and monotone functions are integrable, so upper and lower sums rarely need checking.

Theorem 5.6 (5.2.1)

Every continuous $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. f is uniformly continuous (Theorem 3.18): choose δ with $|f(x) - f(y)| < \epsilon/(b-a)$ when $|x - y| < \delta$. If $\|P\| < \delta$, then $M_i - m_i < \epsilon/(b-a)$ (the extrema are attained), so $U(f, P) - L(f, P) < \epsilon$. ■

Theorem 5.7 (5.2.2)

Every monotone $f : [a, b] \to \mathbb{R}$ is integrable.

Proof. Let f increase and P divide $[a, b]$ into n equal parts. Then $M_i = f(x_i)$, $m_i = f(x_{i-1})$, and $U(f, P) - L(f, P) = \frac{b-a}{n} \sum (f(x_i) - f(x_{i-1})) = \frac{b-a}{n}(f(b) - f(a)) < \epsilon$ for large n. ■

Bounded functions with finitely many discontinuities, or a null discontinuity set (Chapter 10, second volume), are integrable too.

A **Riemann sum** is $S(f,P)=\sum f(t_i)\Delta x_i$ with $t_i\in[x_{i-1},x_i]$.

Theorem 5.8 (5.2.3)

For bounded f and $A\in\mathbb{R}$ the following are equivalent:
(a) $f\in\mathcal{R}$ and $\int_a^b f=A$;
(b) for every $\epsilon>0$ there is $\delta>0$ with $|S(f,P)-A|<\epsilon$ for every partition with $\|P\|<\delta$ and every choice of the t_i.

Proof. (a)⇒(b): $K=\sup|f|$; take P_0 (k points) with $U(f,P_0)-L(f,P_0)<\epsilon/2$, and $\|P\|<\delta=\epsilon/(8Kk)$. Passing to $P\cup P_0$ alters at most k subintervals of P, each changing U or L by $\le 2K\|P\|$, so $U(f,P)-L(f,P)\le U(f,P\cup P_0)-L(f,P\cup P_0)+4Kk\delta<\epsilon$. Both $S(f,P)$ and A lie in $[L(f,P),U(f,P)]$.

(b)⇒(a): for $\|P\|<\delta$, tags with $f(t_i)>M_i-\epsilon/(b-a)$ give $U(f,P)<S(f,P)+\epsilon<A+2\epsilon$; similarly $L(f,P)>A-2\epsilon$. So $U-L<4\epsilon$ and $\int f=A$. ∎

Example 5.2

$\int_0^1 x^2\,dx=\lim_n\frac{1}{n}\sum_{k=1}^n(k/n)^2=\lim\frac{(n+1)(2n+1)}{6n^2}=\frac{1}{3}$. Also $\lim_n\sum_{k=1}^n\frac{1}{n+k}=\int_0^1\frac{dx}{1+x}=\log 2$.

5.3 The fundamental theorem of calculus

Motivation. Differentiation and integration are inverse operations.

For $f\in\mathcal{R}[a,b]$ let $F(x)=\int_a^x f$.

Theorem 5.9 (5.3.1)

F is uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 and $F'(x_0)=f(x_0)$.

Proof. With $K=\sup|f|$, $|F(y)-F(x)|=|\int_x^y f|\le K|y-x|$. If f is continuous at x_0, given ϵ choose δ with $|f(t)-f(x_0)|<\epsilon$ for $|t-x_0|<\delta$; then for $0<|h|<\delta$,

$$\left|\frac{F(x_0+h)-F(x_0)}{h}-f(x_0)\right|=\left|\frac{1}{h}\int_{x_0}^{x_0+h}(f(t)-f(x_0))\,dt\right|\le\epsilon.$$

∎

Corollary 5.10 (5.3.2)

Every continuous $f:[a,b]\to\mathbb{R}$ has a primitive, namely $F(x)=\int_a^x f$.

Theorem 5.11 (5.3.3, fundamental theorem of calculus)

If $F:[a,b]\to\mathbb{R}$ is differentiable and $f=F'$ is integrable, then $\int_a^b f=F(b)-F(a)$.

Proof. For any partition, the mean value theorem gives $t_i\in(x_{i-1},x_i)$ with $F(x_i)-F(x_{i-1})=f(t_i)\Delta x_i$, so $F(b)-F(a)=\sum f(t_i)\Delta x_i\in[L(f,P),U(f,P)]$. As $f\in\mathcal{R}$, $F(b)-F(a)=\int f$. ∎

Integrability of F' is needed: Volterra's F' is bounded but not integrable, and $(x^2\sin(1/x^2))'$ is unbounded on $[0,1]$.

Proposition 5.12 (5.3.4, change of variables)

If $g:[a,b]\to[c,d]$ is a surjective C^1 function and $f:[c,d]\to\mathbb{R}$ is continuous, then $\int_{g(a)}^{g(b)} f(x)\,dx=\int_a^b f(g(t))g'(t)\,dt$.

Proof. With F a primitive of f (Corollary 5.10), $(F\circ g)'=(f\circ g)g'$ is continuous, so both sides equal $F(g(b))-F(g(a))$ (Theorem 5.11). ∎

Proposition 5.13 (5.3.5, integration by parts)

If $F,G:[a,b]\to\mathbb{R}$ are differentiable and $f=F'$, $g=G'$ are integrable, then $\int_a^b Fg = F(b)G(b)-F(a)G(a)-\int_a^b fG$.

Proof. F,G are continuous, hence integrable, so $(FG)'=fG+Fg$ is integrable (products of integrable functions); apply Theorem 5.11 to FG. ∎

5.4 Functions of bounded variation

Motivation. Differences of increasing functions are exactly those of finite total variation.

For $f:[a,b]\to\mathbb{R}$ and a partition P, let $V(f,P)=\sum|f(x_i)-f(x_{i-1})|$. The **total variation** is $V_a^b(f)=\sup_P V(f,P)$; f is of **bounded variation** if it is finite. Refining P increases $V(f,P)$ (triangle inequality). Monotone functions have $V_a^b(f)=|f(b)-f(a)|$; Lipschitz functions with constant K have $V_a^b(f)\le K(b-a)$; BV functions are bounded, $|f(x)|\le|f(a)|+V_a^b(f)$.

Proposition 5.14 (5.4.1)

If f,g are of bounded variation, then so are $f+g$ and fg, with

$$V(f+g)\le V(f)+V(g),\qquad V(fg)\le\|f\|_\infty V(g)+\|g\|_\infty V(f).$$

If moreover $|f|\ge c>0$, then $1/f$ is of bounded variation with $V(1/f)\le V(f)/c^2$.

Proof. Termwise, $|(f+g)(x_i)-(f+g)(x_{i-1})|\le|\Delta f_i|+|\Delta g_i|$, and

$$|f(x_i)g(x_i)-f(x_{i-1})g(x_{i-1})|\le|f(x_i)||\Delta g_i|+|g(x_{i-1})||\Delta f_i|,$$

and $|1/f(x_i)-1/f(x_{i-1})|=|\Delta f_i|/|f(x_i)f(x_{i-1})|\le|\Delta f_i|/c^2$. Sum and take suprema. ∎

Theorem 5.15 (5.4.2)

If f is of bounded variation on $[a,b]$ and $a<c<b$, then f is of bounded variation on $[a,c]$ and $[c,b]$, and $V_a^b(f)=V_a^c(f)+V_c^b(f)$. Conversely, bounded variation on both halves implies it on $[a,b]$.

Proof. For partitions $P\ni c$ of $[a,b]$, $V(f,P)=V(f,P\cap[a,c])+V(f,P\cap[c,b])$, and every partition may be refined to contain c. Taking suprema gives the formula, both sides being finite or infinite together. ∎

Theorem 5.16 (5.4.3, Jordan decomposition)

$f:[a,b]\to\mathbb{R}$ is of bounded variation iff $f=g-h$ with g,h increasing.

Proof. Monotone functions are of bounded variation, hence so are their differences (Proposition 5.14). Conversely, let $g(x)=V_a^x(f)$ (with $g(a)=0$) and $h=g-f$. g increases by Theorem 5.15, and for $x<y$, $h(y)-h(x)=V_x^y(f)-(f(y)-f(x))\geq 0$. ■

Consequently a BV function has one-sided limits everywhere and at most countably many discontinuities (Theorem 3.21, Corollary 3.22). A continuous function need not be of bounded variation: $x\sin(1/x)$ on $[0,1]$.

A **curve** is a continuous $\alpha:[a,b]\to\mathbb{R}^m$; its **length** is $\Lambda(\alpha)=\sup_P\sum\|\alpha(x_i)-\alpha(x_{i-1})\|$.

Proposition 5.17 (5.4.4)

For a curve $\alpha=(\alpha_1,\dots,\alpha_m)$ the following are equivalent:
(a) $\Lambda(\alpha)<\infty$;
(b) each α_k is of bounded variation.

Proof. For each k, $|\alpha_k(x_i)-\alpha_k(x_{i-1})|\leq\|\alpha(x_i)-\alpha(x_{i-1})\|\leq\sum_j|\alpha_j(x_i)-\alpha_j(x_{i-1})|$. Summing, $V(\alpha_k,P)\leq\sum_i\|\Delta\alpha_i\|\leq\sum_j V(\alpha_j,P)$; take suprema. ■

Theorem 5.18 (5.4.5)

Every C^1 curve $\alpha:[a,b]\to\mathbb{R}^m$ has finite length $\Lambda(\alpha)=\int_a^b\|\alpha'(t)\|\,dt$.

Proof. For vector v, $u=\int v$ gives $\|u\|^2=\int\langle u,v\rangle\leq\|u\|\int\|v\|$ (Cauchy–Schwarz), so $\|\alpha(x_i)-\alpha(x_{i-1})\|\leq\int_{x_{i-1}}^{x_i}\|\alpha'\|$ and $\Lambda(\alpha)\leq\int\|\alpha'\|$. Conversely, by uniform continuity $\|\alpha'(t)-\alpha'(x_i)\|<\epsilon$ on $[x_{i-1},x_i]$ for small $\|P\|$, so

$$\int_{x_{i-1}}^{x_i}\|\alpha'\|\leq\|\alpha'(x_i)\|\Delta x_i+\epsilon\Delta x_i=\Big\|\int_{x_{i-1}}^{x_i}(\alpha'(t)+\alpha'(x_i)-\alpha'(t))dt\Big\|+\epsilon\Delta x_i\leq\|\Delta\alpha_i\|+2\epsilon\Delta x_i.$$

Summing, $\int\|\alpha'\|\leq\Lambda(\alpha)+2\epsilon(b-a)$. ■

5.5 The Riemann–Stieltjes integral

Motivation. Replacing Δx by $\Delta\alpha$ makes sums and ordinary integrals special cases of one integral.

Let $\alpha:[a,b]\to\mathbb{R}$ be increasing. With $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})$ define $U(f,P,\alpha)=\sum M_i\Delta\alpha_i$, $L(f,P,\alpha)=\sum m_i\Delta\alpha_i$, and $S(f,P,\alpha)=\sum f(t_i)\Delta\alpha_i$. As in Proposition 5.1, refinement decreases U and increases L. $f\in\mathcal{R}(\alpha)$ if $\inf_P U(f,P,\alpha)=\sup_P L(f,P,\alpha)$; the common value is $\int_a^b f\,d\alpha$.

Theorem 5.19 (5.5.1)

Let α,β be increasing and f,g bounded.
(a) If $f,g\in\mathcal{R}(\alpha)$ and $c\in\mathbb{R}$, then $f+g,cf\in\mathcal{R}(\alpha)$ with $\int(f+g)\,d\alpha=\int f\,d\alpha+\int g\,d\alpha$ and $\int cf\,d\alpha=c\int f\,d\alpha$; and $f\leq g$ implies $\int f\,d\alpha\leq\int g\,d\alpha$.
(b) If $f\in\mathcal{R}(\alpha)$ on $[a,b]$ and $a<c<b$, then $f\in\mathcal{R}(\alpha)$ on $[a,c]$ and $[c,b]$, and $\int_a^b=\int_a^c+\int_c^b$.
(c) If $f\in\mathcal{R}(\alpha)\cap\mathcal{R}(\beta)$ and $c\geq 0$, then $f\in\mathcal{R}(\alpha+\beta)\cap\mathcal{R}(c\alpha)$ with $\int f\,d(\alpha+\beta)=\int f\,d\alpha+\int f\,d\beta$ and $\int f\,d(c\alpha)=c\int f\,d\alpha$.
(d) If $f\in\mathcal{R}(\alpha)$, then $|f|\in\mathcal{R}(\alpha)$ and $|\int f\,d\alpha|\leq\int|f|\,d\alpha\leq\sup|f|\,(\alpha(b)-\alpha(a))$.

Proof. The proofs of Theorems 5.3, 5.4 and 5.5 apply verbatim with Δx_i replaced by $\Delta\alpha_i \ge 0$. For (c), $U(f,P,\alpha+\beta) = U(f,P,\alpha) + U(f,P,\beta)$ and likewise for L. ∎

Theorem 5.20 (5.5.2)

For bounded f and increasing α the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$;
(b) for every $\epsilon > 0$ there is P with $U(f,P,\alpha) - L(f,P,\alpha) < \epsilon$;
(c) there is A such that for every $\epsilon > 0$ some P_0 has $|S(f,P,\alpha) - A| < \epsilon$ for every $P \supset P_0$ and every choice of tags. In that case $A = \int f\,d\alpha$.

Proof. (a)⇔(b) as in Proposition 5.2.

(b)⇒(c): let $A = \int f\,d\alpha$ and P_0 be as in (b). For $P \supset P_0$, both $S(f,P,\alpha)$ and A lie in the interval $[L(f,P,\alpha), U(f,P,\alpha)]$ of length $< \epsilon$.

(c)⇒(b): for $P \supset P_0$, taking suprema and infima over tags, $U(f,P,\alpha) \le A+\epsilon$ and $L(f,P,\alpha) \ge A-\epsilon$. ∎

Example 5.3

(a) If α is the unit step at $c \in (a,b)$ ($\alpha = 0$ on $[a,c]$, 1 on $(c,b]$) and f is continuous at c, then $\int f\,d\alpha = f(c)$. If f and α are both discontinuous from the right at c, f may fail to be integrable.
(b) For a step function α with jumps c_k at points s_k and continuous f, $\int f\,d\alpha = \sum c_k f(s_k)$: sums are a special case of Stieltjes integrals.

For α of bounded variation write $\alpha = \alpha_1 - \alpha_2$, α_i increasing (Theorem 5.16); if $f \in \mathcal{R}(\alpha_1) \cap \mathcal{R}(\alpha_2)$, then $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = \int f\,d\alpha_1 - \int f\,d\alpha_2$, independent of the decomposition by Theorem 5.22. $S(f,P,\alpha)$ is as before.

Proposition 5.21 (5.5.3)

If f is continuous on $[a,b]$, then $f \in \mathcal{R}(\alpha)$ for every α of bounded variation.

Proof. It suffices to treat increasing α. Given ϵ, by uniform continuity choose P with $M_i - m_i < \epsilon$ for all i; then $U(f,P,\alpha) - L(f,P,\alpha) \le \epsilon(\alpha(b) - \alpha(a))$. ∎

Theorem 5.22 (5.5.4)

For bounded f and α of bounded variation the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = A$;
(b) for every $\epsilon > 0$ there is P_0 with $|S(f,P,\alpha) - A| < \epsilon$ for all $P \supset P_0$ and all tags.

Proof. (a)⇒(b): $S(f,P,\alpha) = S(f,P,\alpha_1) - S(f,P,\alpha_2)$; apply Theorem 5.20(c) to each and take a common refinement. In particular A is determined by f and α, so the integral is well defined.

(b)⇒(a): let $V(x) = V_a^x(\alpha)$ and $K = \sup|f|$; we show $f \in \mathcal{R}(V)$. Refine P_0 from (b) so that $V(b) - \sum_i |\Delta\alpha_i| < \epsilon$ (refining keeps this). For $P \supset P_0$,

$$U(f,P,V) - L(f,P,V) = \sum (M_i - m_i)(\Delta V_i - |\Delta\alpha_i|) + \sum (M_i - m_i)|\Delta\alpha_i|.$$

As $\Delta V_i \ge |\Delta\alpha_i|$, the first sum is $\le 2K\epsilon$. For the second, pick s_i, t_i in the i-th interval with $f(t_i) - f(s_i) > M_i - m_i - \epsilon$, swapped where $\Delta\alpha_i < 0$; it is then $< \sum (f(t_i) - f(s_i))\Delta\alpha_i + \epsilon V(b) = S(f,P,\alpha) - S'(f,P,\alpha) + \epsilon V(b) < 2\epsilon + \epsilon V(b)$ (S, S' with tags t_i, s_i). So $f \in \mathcal{R}(V)$ (Theorem 5.20).

Then $S(f,P,V-\alpha) = S(f,P,V) - S(f,P,\alpha)$ satisfies Theorem 5.20(c) for the increasing $V-\alpha$, so $f \in \mathcal{R}(V-\alpha)$, and $\alpha = V - (V-\alpha)$ gives $\int f\,d\alpha = A$. ■

Theorem 5.23 (5.5.5)

Let α be differentiable with α' continuous on $[a,b]$. If f is bounded and $f \in \mathcal{R}(\alpha)$, then $f\alpha'$ is Riemann integrable and $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. α is of bounded variation (Lipschitz). By the mean value theorem $\Delta\alpha_i = \alpha'(u_i)\Delta x_i$ with $u_i \in (x_{i-1}, x_i)$, so

$$\left|S(f,P,\alpha) - \sum f(t_i)\alpha'(t_i)\Delta x_i\right| \le K\sum |\alpha'(u_i) - \alpha'(t_i)|\Delta x_i.$$

This is $\le K\epsilon(b-a)$ for small $\|P\|$ (uniform continuity of α'). With Theorem 5.22(b), Riemann sums of $f\alpha'$ over fine refinements of some P_0 lie within $\epsilon + K\epsilon(b-a)$ of $\int f\,d\alpha$; apply Theorem 5.20 with $\alpha(x) = x$. ■

Theorem 5.24 (5.5.6, integration by parts)

Let f, α be of bounded variation on $[a,b]$. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ and $\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a)$.

Proof. Let $A = \int f\,d\alpha$ and P_0 as in Theorem 5.22(b). Let $Q \supset P_0$ be a partition $a = y_0 < \cdots < y_m = b$ with tags $s_k \in [y_{k-1}, y_k]$. By summation by parts,

$$S(\alpha,Q,f) = \sum_k \alpha(s_k)(f(y_k) - f(y_{k-1})) = f(b)\alpha(b) - f(a)\alpha(a) - \sum_{k=0}^{m} f(y_k)(\alpha(s_{k+1}) - \alpha(s_k)),$$

with $s_0 = a$, $s_{m+1} = b$. The last sum is $S(f,P,\alpha)$ for the partition $P \supset Q \supset P_0$ of all s_k and y_k (zero-length intervals contribute nothing), tags y_k. So $|S(\alpha,Q,f) - (f(b)\alpha(b) - f(a)\alpha(a) - A)| < \epsilon$; apply Theorem 5.22. ■

Corollary 5.25 (5.5.7, second mean value theorem)

If $f : [a,b] \to \mathbb{R}$ is increasing and g is continuous, there is $c \in [a,b]$ with $\int_a^b fg = f(a)\int_a^c g + f(b)\int_c^b g$.

Proof. $G(x) = \int_a^x g$ is C^1, $G' = g$. $G \in \mathcal{R}(f)$ (Proposition 5.21), so $\int fg = \int f\,dG = f(b)G(b) - \int G\,df$ (Theorems 5.23, 5.24). As $\min G \cdot (f(b) - f(a)) \le \int G\,df \le \max G \cdot (f(b) - f(a))$, the intermediate value theorem gives $\int G\,df = G(c)(f(b) - f(a))$. Hence $\int fg = f(b)G(b) - G(c)(f(b) - f(a)) = f(a)G(c) + f(b)(G(b) - G(c))$. ■

Appendix A. The Number Systems

We construct a complete ordered field from $\mathbb{N} = \{0, 1, 2, \dots\}$ (with well-ordering): $\mathbb{Z}$ and $\mathbb{Q}$ by equivalence relations, $\mathbb{R}$ from Cauchy sequences of rationals.

A.1 Integers and rationals

Motivation. First, how $\mathbb{Z}$ and $\mathbb{Q}$ arise from $\mathbb{N}$.

On $\mathbb{N} \times \mathbb{N}$ put $(m, n) \sim (m', n') \iff m + n' = n + m'$ (think of $m - n$); this is an equivalence relation, and $(m, n) \sim (m - k, n - k)$ when $m, n \geq k$. $\mathbb{Z}$ is the set of classes $[m, n]$, with $[m, n] \geq [k, l] \iff m + l \geq n + k$, $[m, n] + [k, l] = [m + k, n + l]$, $[m, n] \cdot [k, l] = [mk + nl, ml + nk]$. These are well defined (independent of representatives), and every integer is $[n, 0]$, $[0, 0]$ or $[0, n]$ for a unique $n \geq 1$, written n, 0, $-n$; the map $n \mapsto [n, 0]$ embeds $\mathbb{N}$ preserving operations and order.

Theorem A.1 (Theorem 1)

(a) For $a, b \in \mathbb{Z}$, $a \geq b$ or $b \geq a$.
(b) A nonempty subset of $\mathbb{Z}$ bounded above has a largest element, and one bounded below has a least element.

Proof. (a) reduces to the same property of $\mathbb{N}$ for $m + l$ and $n + k$.

(b) If S is bounded below by b, then $\{s - b : s \in S\} \subset \mathbb{N}$ has a least element k, and $b + k$ is least in S; for S bounded above apply this to $-S$. ∎

$\mathbb{Z}$ satisfies the field axioms except the existence of multiplicative inverses, and with $P_{\mathbb{Z}} = \{1, 2, \dots\}$ the order axioms. On $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ put $(a, b) \sim (c, d) \iff ad = bc$ and order the classes by $(a, b) > (c, d) \iff abd^2 > cdb^2$; the classes are the rationals a/b, with $a/b + c/d = (ad + bc)/bd$ and $(a/b)(c/d) = ac/bd$. One checks that $\mathbb{Q}$ is an ordered field containing $\mathbb{Z}$, so the results of Section 1.1, valid in any ordered field, hold in $\mathbb{Q}$.

A.2 Real numbers as classes of Cauchy sequences

Motivation. A real number is the class of the rational Cauchy sequences that "should" converge to it.

A rational sequence $(a(i))$ is a **fundamental sequence** if for every rational $\epsilon > 0$ there is N with $|a(i) - a(j)| < \epsilon$ for $i, j \geq N$.

Theorem A.2 (Theorem 2)

(a) Fundamental sequences are bounded.
(b) A bounded monotone rational sequence is fundamental.
(c) If $(a(i))$ is fundamental, so is $(|a(i)|)$.
(d) If $(a(i)), (b(i))$ are fundamental, so are $(a(i) \pm b(i))$ and $(a(i)b(i))$.
(e) If $(a(i))$ is fundamental and there are N and a rational $d > 0$ with $|a(i)| \geq d$ for $i \geq N$, then $(1/a(i))_{i \geq N}$ is fundamental.

Proof. (a) as in Proposition 2.12.

(b) If an increasing sequence bounded by B were not fundamental, there would be a rational $\epsilon > 0$ and indices $i_1 < j_1 \leq i_2 < j_2 \leq \cdots$ with $a(j_k) - a(i_k) \geq \epsilon$; then $a(j_k) \geq a(i_1) + k\epsilon > B$ for large k.

(c) $||a(i)| - |a(j)|| \le |a(i) - a(j)|$.

(d) $|a(i)b(i) - a(j)b(j)| \le |a(i)||b(i) - b(j)| + |b(j)||a(i) - a(j)|$ and (a).

(e) $|1/a(i) - 1/a(j)| \le |a(i) - a(j)|/d^2$. ■

Two fundamental sequences are **equivalent**, $(a(i)) \sim (b(i))$, if for every rational $\epsilon > 0$ there is N with $|a(i) - b(i)| < \epsilon$ for $i \ge N$; this is an equivalence relation. A **real number** is an equivalence class $[a]$; $\mathbb{R}$ is the set of classes. Define $[a] > [b]$ if there are a rational $d > 0$ and N with $a(i) - b(i) > d$ for $i \ge N$; this does not depend on the representatives.

Theorem A.3 (Theorem 3, trichotomy)

For $[a], [b] \in \mathbb{R}$ exactly one of $[a] > [b]$, $[a] = [b]$, $[b] > [a]$ holds.

Proof. For rational $\epsilon > 0$ let N_ϵ be such that $|a(i) - a(N_\epsilon)| < \epsilon$ and $|b(i) - b(N_\epsilon)| < \epsilon$ for $i \ge N_\epsilon$. Then for $i \ge N_\epsilon$,

$$d_\epsilon := a(N_\epsilon) - b(N_\epsilon) - 2\epsilon < a(i) - b(i) < a(N_\epsilon) - b(N_\epsilon) + 2\epsilon =: d'_\epsilon.$$

If $d_\epsilon > 0$ for some ϵ, then $[a] > [b]$; if $d'_\epsilon < 0$ for some ϵ, then $[b] > [a]$. Otherwise for every ϵ, $|a(N_\epsilon) - b(N_\epsilon)| \le 2\epsilon$, so $|a(i) - b(i)| < 4\epsilon$ for $i \ge N_\epsilon$, and $[a] = [b]$. Two of the cases cannot hold together, by the definitions. ■

Define $[a]+[b] = [a+b]$ and $[a][b] = [ab]$ (termwise operations); these are well defined by Theorem A.2. For $r \in \mathbb{Q}$ let r^* be the class of the constant sequence r; $r \mapsto r^*$ embeds $\mathbb{Q}$ in $\mathbb{R}$ preserving operations and order. 0^* and 1^* are the identities, $[-a]$ is the additive inverse, and if $[a] \ne 0^*$ then by Theorem A.3 there are $d > 0$ and N with $|a(i)| \ge d$ for $i \ge N$, so the class of $(1/a(i))_{i \ge N}$ (Theorem A.2(e)) is the inverse. With $P = \{[a] : [a] > 0^*\}$, $\mathbb{R}$ is an ordered field, and $|[a]| = [|a|]$.

Theorem A.4 (Theorem 4, density of $\mathbb{Q}$)

If $[b] > [a]$, there is a rational r with $[b] > r^* > [a]$.

Proof. Choose N_1 and rational $d > 0$ with $b(i) - a(i) > d$ for $i \ge N_1$, and N_2 with $|a(i) - a(j)| < d/4$, $|b(i) - b(j)| < d/4$ for $i, j \ge N_2$; let $N = \max(N_1, N_2)$ and $r = a(N) + d/2$. For $i \ge N$, $r - a(i) = d/2 + (a(N) - a(i)) > d/4$, so $r^* > [a]$; and $b(i) - r = (b(i) - b(N)) + (b(N) - a(N)) - d/2 > -d/4 + d - d/2 = d/4$, so $[b] > r^*$. ■

Theorem A.5 (Theorem 5, completeness)

Every Cauchy sequence $([a_n])$ in $\mathbb{R}$ converges in $\mathbb{R}$.

Proof. If the sequence is eventually constant there is nothing to prove; otherwise, passing to a subsequence (which suffices by Lemma 2.13, whose proof is valid in any ordered field), assume $[a_n] \ne [a_{n+1}]$ for all n. By Theorem A.4 choose rationals $r(n)$ strictly between $[a_n]$ and $[a_{n+1}]$; then $|[a_n] - r(n)^*| < |[a_n] - [a_{n+1}]| \to 0$. The rational sequence $(r(n))$ is fundamental, since $|r(n) - r(m)|^* \le |r(n)^* - [a_n]| + |[a_n] - [a_m]| + |[a_m] - r(m)^*|$. Let $[r]$ be its class. Given $[e] > 0$ choose, by Theorem A.4, a rational $\epsilon > 0$ with $\epsilon^* < [e]$, and M with $|r(i) - r(j)| < \epsilon/2$ for $i, j \ge M$. For $n \ge M$, $|r(n)^* - [r]| \le (\epsilon/2)^*$, because the termwise differences $r(n) - r(i)$ are eventually below $\epsilon/2$. Hence for large n, $|[a_n] - [r]| \le |[a_n] - r(n)^*| + (\epsilon/2)^* < [e]$. ■

Theorems A.4 and A.5 give the completeness axiom: bisection (after Theorem 1.20) yields nested rational intervals with lengths $\to 0$, whose left ends are Cauchy with limit $\sup S$. So $\mathbb{R}$ is a complete ordered field, unique up to isomorphism (Theorem 1.10).

A.3 Selected exercises

Motivation. Two checks the text omits: independence of representatives, and $\sqrt{2} \in \mathbb{R}$.

Throughout, $a, b, \ldots$ denote fundamental sequences of rationals, and ϵ, d rational numbers.

Exercise A.1 (Well-definedness)

(a) Show that $\sim$ is an equivalence relation on fundamental sequences.
(b) Show that if $a \sim a'$ and $b \sim b'$, then $ab \sim a'b'$.
(c) Show that if $a \sim a'$, $b \sim b'$ and $[a] > [b]$ via a, b, then also $a'(i) - b'(i) > d'$ for some rational $d' > 0$ and all large i.

Solution [Supplement].

(a) Reflexivity and symmetry are clear. Transitivity: if $|a(i) - b(i)|, |b(i) - c(i)| < \epsilon/2$ for $i \geq N$, then $|a(i) - c(i)| < \epsilon$.

(b)

Step 1. Let $|a(i)|, |b'(i)| \leq B$ (Theorem A.2(a)). Then

$$|a(i)b(i) - a'(i)b'(i)| \leq |a(i)|\,|b(i) - b'(i)| + |b'(i)|\,|a(i) - a'(i)|.$$

Step 2. Once $|b(i) - b'(i)|, |a(i) - a'(i)| < \epsilon/(2B)$ the left side is $< \epsilon$; both products are fundamental (Theorem A.2(d)), so $ab \sim a'b'$.

(c) Let $a(i) - b(i) > d$ for $i \geq N_1$ and $|a(i) - a'(i)|, |b(i) - b'(i)| < d/4$ for $i \geq N_2$. For $i \geq \max(N_1, N_2)$,

$$a'(i) - b'(i) \geq (a(i) - b(i)) - |a(i) - a'(i)| - |b(i) - b'(i)| > d - \tfrac{d}{2} = \tfrac{d}{2},$$

so $d' = d/2$ works. ∎

Exercise A.2 (A square root of 2)

Define rationals $a(0) = 2$ and $a(i+1) = \frac{a(i)}{2} + \frac{1}{a(i)}$.
(a) Show that $a(i) > 0$, $a(i)^2 \geq 2$ and $a(i+1) \leq a(i)$ for all i; conclude that $(a(i))$ is fundamental.
(b) Show that $0 \leq a(i)^2 - 2 \leq 2 \cdot 4^{-i}$, and deduce that $x = [a]$ satisfies $x > 0^*$ and $x^2 = 2^*$.

Solution [Supplement].

(a)

Step 1. $a(i) > 0 \Rightarrow a(i+1) > 0$, so all $a(i)$ are positive rationals.

Step 2.

$$a(i+1)^2 - 2 = \frac{a(i)^2}{4} + 1 + \frac{1}{a(i)^2} - 2 = \Big(\frac{a(i)}{2} - \frac{1}{a(i)}\Big)^2 = \frac{(a(i)^2 - 2)^2}{4a(i)^2} \geq 0.$$

With $a(0)^2 = 4$, $a(i)^2 \geq 2$ for all i.

Step 3. $a(i) - a(i+1) = \frac{a(i)^2 - 2}{2a(i)} \geq 0$: $(a(i))$ decreases and is bounded below by 1.

Step 4. So $(-a(i))$, hence $(a(i))$, is fundamental (Theorem A.2(b)).

(b)

Step 1. $e(i) = a(i)^2 - 2 \geq 0$ satisfies $e(i+1) = e(i)^2/(4a(i)^2) \leq e(i)^2/8$.

Step 2. $e(0) = 2$, and $e(i) \leq 2 \cdot 4^{-i} \leq 2$ gives $e(i+1) \leq \frac{e(i)}{4} \leq 2 \cdot 4^{-(i+1)}$; induct.

Step 3. So $|a(i)a(i) - 2| < \epsilon$ eventually: $x^2 = [aa] = 2^*$.

Step 4. $a(i) \geq 1 > \frac{1}{2}$, so $x > 0^*$ $(d = \frac{1}{2})$.

As no rational squares to 2, $(a(i))$ has no limit in $\mathbb{Q}$: the incompleteness the construction repairs. ■